COMPUTATIONAL IDENTIFICATION AND THERAPEUTIC TARGETING OF NUSAP1 IN BREAST CANCER: PATHOGENIC nsSNPs CHARACTERIZATION, STRUCTURAL ANALYSIS, AND DUAL VIRTUAL SCREENING OF CHEMICAL AND PHYTOCHEMICAL COMPOUNDS

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ABSTRACT

Breast Cancer is the leading cause of cancer-related death in women. A total of 87 hub genes were identified through a systematic review of published bioinformatics studies. Identification of therapeutic drug targets using the Therapeutic Targets Database (TTD) for the reported 87 hub genes; construction of protein-protein interactions using STRING; and identification of hub genes lacking therapeutic drug target type and drug names using the CytoHubba plugin in Cytoscape. Hub Gene expression for the top 10 ranking hub genes (18) in breast cancer and pan-cancer using GEPIA2. NUSAP1 was selected for further analysis as it shows high expression in breast cancer and pan-cancer analysis, and it also lacks a therapeutic drug target type and drug name. 460 Variants for the NUSAP1 protein were retrieved from UniProtKB, and three deleterious non-synonymous SNPs(nsSNPs) -N386S, K404E, and R434M were identified using the Ensembl Variant Effect Predictor with prediction tools such as SIFT, PolyPhen2, MutationTaster, PROVEAN, MutationAssessor, MetaSVM, and ClinPred, and the T114N variant was selected as it was reported in UniProtKB as a cancer-associated variant. Protein stability analysis for four variants was performed using I-Mutant 2.0 and MUpro, which destabilizes the NUSAP1 protein structure. Structural effects were evaluated using HOPE, and conserved residues were identified using the Evolutionary Trace (ET) database. 3D wild-type NUSAP1, R434M, and T114N models were generated using AlphaFold, then refined using GalaxyRefine, and their quality was validated on SAVES. Normal mode analysis determined that the native dynamics of the WT protein are most conserved, and it has the optimal stability-to-flexibility ratio compared to the R434M variant and the T114N variant, the least stable structure (WT > R434M > T114N). Wildtype was analyzed for binding pockets, using PrankWeb, and then virtually screened against the Mcule compound library and ~17,000 phytocompounds (IMPPAT 2.0 database) using AutoDock Vina. The best docking poses (lead compounds) were then filtered based on Lipinski's Rule of Five, assessed for drug-likeness using SwissADME, and for toxicity using ProTox 3.0.

Keywords:

Cancer, Breast cancer, NUSAP1 gene, STRING, Cytoscape, Single nucleotide Polymorphisms (SNPs), Deleterious Single nucleotide polymorphisms(dsSNPs), Molecular docking.

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ABBREVIATIONS

ACC – Adrenocortical Carcinoma

BLCA – Bladder Urothelial Carcinoma

BRCA – Breast Invasive Carcinoma

CESC – Cervical Squamous Cell Carcinoma and Endocervical Adenocarcinoma

CHOL – Cholangiocarcinoma

COAD – Colon Adenocarcinoma

DLBC – Diffuse Large B-cell Lymphoma

ESCA – Esophageal Carcinoma

GBM – Glioblastoma Multiforme

KICH – Kidney Chromophobe

KIRC – Kidney Renal Clear Cell Carcinoma

KIRP – Kidney Renal Papillary Cell Carcinoma

LAML – Acute Myeloid Leukemia

LGG – Brain Lower Grade Glioma

LIHC – Liver Hepatocellular Carcinoma

LUAD – Lung Adenocarcinoma

LUSC – Lung Squamous Cell Carcinoma

MESO – Mesothelioma

OV – Ovarian Serous Cystadenocarcinoma

PAAD – Pancreatic Adenocarcinoma

PCPG – Pheochromocytoma and Paraganglioma

PRAD – Prostate Adenocarcinoma

READ – Rectum Adenocarcinoma

SARC – Sarcoma

SKCM – Skin Cutaneous Melanoma

STAD – Stomach Adenocarcinoma

TGCT – Testicular Germ Cell Tumors

THCA – Thyroid Carcinoma

THYM – Thymoma

UCEC – Uterine Corpus Endometrial Carcinoma

UCS – Uterine Carcinosarcoma

UVM – Uveal Melanoma.

CCNB1 – Cyclin B1

RRM2 – Ribonucleotide Reductase Regulatory Subunit M2

TOP2A – DNA Topoisomerase II Alpha

CDK1 – Cyclin-Dependent Kinase 1

AURKA – Aurora Kinase A

BUB1B – BUB1 Mitotic Checkpoint Serine/Threonine Kinase B

NUSAP1 – Nucleolar and Spindle Associated Protein 1

PPARG – Peroxisome Proliferator-Activated Receptor Gamma

KIF11 – Kinesin Family Member 11

CCNA2 – Cyclin A2

CCNB2 – Cyclin B2

BIRC5 – Baculoviral IAP Repeat Containing 5

MELK – Maternal Embryonic Leucine Zipper Kinase

DTL – Denticleless E3 Ubiquitin Protein Ligase Homolog

PRC1 – Protein Regulator of Cytokinesis 1

TPX2 – TPX2 Microtubule Nucleation Factor

KIF20A – Kinesin Family Member 20A

MAD2L1 – Mitotic Arrest Deficient 2 Like 1

CENPF – Centromere Protein F

CEP55 – Centrosomal Protein 55

LEP – Leptin

FGF2 – Fibroblast Growth Factor 2

PBK – PDZ Binding Kinase

KIF2C – Kinesin Family Member 2C

ASPM – Abnormal Spindle Microtubule Assembly

EZH2 – Enhancer of Zeste 2 Polycomb Repressive Complex 2 Subunit

CDC20 – Cell Division Cycle 20

BUB1 – BUB1 Mitotic Checkpoint Serine/Threonine Kinase

CDCA8 – Cell Division Cycle Associated 8

HMMR – Hyaluronan-Mediated Motility Receptor

KIF23 – Kinesin Family Member 23

UBE2C – Ubiquitin Conjugating Enzyme E2 C

RACGAP1 – Rac GTPase Activating Protein 1

DLGAP5 – DLG Associated Protein 5

IGF1 – Insulin-Like Growth Factor 1

PTEN – Phosphatase and Tensin Homolog

FOXO1 – Forkhead Box O1

PIK3CA – Phosphatidylinositol-4,5-Bisphosphate 3-Kinase Catalytic Subunit Alpha

CDH1 – Cadherin 1

RPS9 – Ribosomal Protein S9

RPL11 – Ribosomal Protein L11

RPS14 – Ribosomal Protein S14

RPL10A – Ribosomal Protein L10a

RRM2B – Ribonucleotide Reductase Regulatory TP53 Inducible Subunit M2B

CDC20B – Cell Division Cycle 20B

MCM4 – Minichromosome Maintenance Complex Component 4

NEK2 – NIMA Related Kinase 2

FABP4 – Fatty Acid Binding Protein 4

PLIN1 – Perilipin 1

ADIPOQ – Adiponectin, C1Q, and Collagen Domain Containing

CD36 – CD36 Molecule

LIPE – Lipase E, Hormone-Sensitive Type

LPL – Lipoprotein Lipase

CIDEA – Cell Death Inducing DFFA Like Effector A

CIDEC – Cell Death Inducing DFFA-Like Effector C

NDC80 – NDC80 Kinetochore Complex Component

CD274 – CD274 Molecule (PD-L1)

KRT5 – Keratin 5

KRT16 – Keratin 16

TP63 – Tumor Protein P63

KRT7 – Keratin 7

CXCL5 – C-X-C Motif Chemokine Ligand 5

HGF – Hepatocyte Growth Factor

TNFRSF8 – TNF Receptor Superfamily Member 8

TNFRSF1B – TNF Receptor Superfamily Member 1B

MMP13 – Matrix Metallopeptidase 13

CTSL – Cathepsin L

CXCL1 – C-X-C Motif Chemokine Ligand 1

CASP1 – Caspase 1

MAP1B – Microtubule Associated Protein 1B

SQSTM1 – Sequestosome 1

IRAK1 – Interleukin 1 Receptor Associated Kinase 1

TLR3 – Toll-Like Receptor 3

PYCARD – PYD And CARD Domain Containing

CASP6 – Caspase 6

CLEC7A – C-Type Lectin Domain Containing 7A

CENPA – Centromere Protein A

PCNA – Proliferating Cell Nuclear Antigen

KIF4A – Kinesin Family Member 4A

KPNA2 – Karyopherin Subunit Alpha 2

TYMS – Thymidylate Synthetase

RRM1 – Ribonucleotide Reductase Catalytic Subunit M1

ZWINT – ZW10 Interacting Kinetochore Protein

NCAPG – Non-SMC Condensin I Complex Subunit G

PLK1 – Polo Like Kinase 1

KIAA0101 – KIAA0101 PCNA Clamp Associated Factor

GINS1 – GINS Complex Subunit 1

BC- Breast Cancer

WT- Wildtype

CHAPTER 1: INTRODUCTION

1.1 Cancer:

Cancer is one of the most common causes of death in the world. Cancer is a disease in which some of the body's cells grow uncontrollably and spread to other parts of the body [1]. Normally, human cells grow and undergo cell division to form new cells; old cells age or get damaged, then die, and new cells replace them. Cancer cells grow uncontrollably and form tumors, which are lumps in tissue. Two types of tumors: cancerous tumors (malignant) and non-cancerous tumors (benign) [2]. Symptoms of Cancer are Fatigue, Fever, Loss of appetite, new lumps or bumps, etc. Cancer is a genetic disorder, which means when genes that manage cell activity change(mutate), they create abnormal cells that divide and grow uncontrollably, which disturbs the body's work [3]. Causes of cancers are family history, smoking, malnutrition, hormones, and radiation exposure, etc. Cancers can be caused in any part of the body, such as the breast, lung, liver, pancreas, thyroid, and kidney, etc. [4].

1.2 Breast Cancer:

Breast cancer is a type of cancer that begins as an uncontrollable growth of cells in the breast, commonly forming lumps called tumors [5]. Breast cancer can form in glandular tissue (it includes milk ducts, milk glands and lobules tiny glands that make milk) cancers that start in the duct are called ductal cancers, cancers that start in lobules are called lobular cancer[6], fibrous tissue and fatty tissue (it fills the space between ducts and lobules, they give breasts their shape and size) cancers that start in fatty tissue called phyllodes tumor, nipple(small, raised area in the center of breast which contains duct openings through which milk can secrete) cancer that start in the nipple called Paget disease and blood vessels and lymph vessels (cancer occur when cancer cells block lymph vessels in the breast) cancer that occur in the line of blood vessels and lymph vessels called Angiosarcoma[7].

1.3 Oncogenes:

A gene is the basic physical and functional unit of heredity. Genes are made of DNA and are passed from parents to offspring; they contain information about physical and biological traits that enable the development, growth, and reproduction of life [8]. A

gene consists of a long sequence of combinations of four different nucleotide bases (Adenine, Cytosine, Guanine, Thymine), which are building blocks of DNA. When the sequence of DNA changes due to insertions, substitutions, deletions, or other alterations, it can lead to the uncontrolled growth of cells, resulting in cancer [9]. Cancer genes are three types of genes that play a key role in cancer: oncogenes, tumor-suppressor genes, and DNA-repair genes. Oncogenes are mutated genes that are involved in the development of cancer. Before becoming oncogenes (damaged cells), they are normal cells(proto-oncogenes). The damaged cells divide uncontrollably, causing cancer. Tumor suppressor genes are a type of cell growth control; they slow down the division of normal cells. When tumor suppressor genes are not working, then the cell growth leads to uncontrollable growth, which leads to cancer [8]. DNA-repair genes, the DNA in every cell in our body is constantly in danger of becoming damaged. But cells contain many proteins, which help repair the damaged DNA. If a cell has less ability to repair itself which can lead to form cancer [10].

1.4 Single Nucleotide Polymorphism (SNPs):

SNP stands for Single Nucleotide Polymorphism, a genomic variant at a single base in DNA. SNPs occur when one nucleotide base is replaced or substituted with another nucleotide base [11]. SNPs are classified based on their location and functional impact they are coding region SNPs (occur within protein-coding genes, they are classified into two types synonymous – don't alter the amino acid sequence and non-synonymous SNPs- alter the amino acid sequence) and non-coding region SNPs occur in 5′untranslated region (UTR), 3′untranslated region, introns, promoters it may affect gene regulation, transcription or mRNA stability [12]. Deleterious Single Nucleotide Polymorphisms (dsSNPs) are single-base changes in the DNA sequence of a gene that can interfere with normal biological processes and are often associated with diseases such as cancer, genetic disorders, etc. The term deleterious SNPs means harmful single-nucleotide polymorphisms. dsSNPs are classified based on how they impact the expression of gene and protein function [13]. Missense SNPs are a type of dsSNPs that are classified based on their effect on protein structure. Missense SNPs are substitutions of one amino acid for another, which may lead to a change in protein structure and function. Missense SNPs can cause cancer, genetic disorders, and metabolic diseases [14].

CHAPTER 2: REVIEW OF LITERATURE

2.1 Molecular classification of Breast Cancer:

Breast cancer (BC) is the most common type of cancer and the second most common cause of cancer-related death in women [15]. Breast cancer originates from the epithelium of the mammary gland [16]. BC is classified into several categories based on different molecular features, histological appearance, and clinical outcomes. Breast cancer is a molecularly heterogeneous disorder [17]. BC can be split into six subgroups based on molecular classifications: normal-like, luminal A and B, HER2-positive, basal-like, and claudin-low. [18]. The identification of novel prognostic biomarkers and the improvement of the available treatment modalities are highly important for breast cancer patients [19]. BC is a highly lethal tumor and ranks second in the most susceptible tumor among women, but the molecular mechanisms of BC occurrence and progression are not fully understood [20]. Understanding the molecular mechanisms of BC and identifying novel potential prognostic biomarkers to improve the prognosis of BC are urgently needed [21]. BC treatment commonly includes surgical resection and hormone therapy, radiotherapy, or chemotherapy. BC remains highly prevalent and malignant due to recurrence and metastasis [22].

2.2 Identification of Differentially expressed genes and hub genes involved in Breast cancer:

After reviewing the research papers, many researchers have used gene expression data from the Gene Expression Omnibus database [26]. The selection of datasets was based on sample size, condition-based, and the availability of disease and control group samples for analysis. Normalization and differential expression analysis were performed using tools such as limma, DESeq2, in R, or GEO2R for the identification of differentially expressed genes. An adjusted p-value < 0.05 and a $\log_2$ fold change ($\log_2$FC) > 1 were common thresholds for the identification of significant genes, and they are visualized using volcano plots and heatmaps to compare expression profiles between disease and control samples. The Venny tool was used to identify the intersection of differentially expressed genes in different datasets. Protein-Protein Interaction was constructed for the commonly identified differentially expressed genes to identify the known or predicted interactions between the common DEGs. Hub genes

were identified from the constructed PPI network in STRING using the Cytohubba plugin in Cytoscape. Based on the Maximal Clique Centrality (MCC) algorithm, hub genes were identified. The top-ranked genes were considered them as hub genes. Survival analysis was performed to examine the association between expression levels of identified hub genes and patient survival. by using Kaplan-Meier (KM) plotter, TNMplot database, and The Cancer Genome Atlas (TCGA). Patients were classified into the high-expression group and the low-expression group based on the median expression of each hub gene, and survival differences were analyzed using the log-rank test. Functional enrichment analysis was performed on hub genes using bioinformatics analysis tools such as Enrichr and DAVID. The Gene Ontology (GO) enrichment analysis was performed to annotate the genes and classify them into three main categories, i.e., Biological Process (BP), Molecular Function (MF), and Cellular Component (CC). Enrichment analysis results were visualized using bar plots, dot plots, or enriched gene network maps

2.3 Reported Hub genes in Breast cancer:

A hub gene is a highly connected and centrally positioned gene in a biological network, such as protein-protein interaction networks. In such networks, genes are represented as nodes and interactions between the genes are represented as edges. Hub genes are different from genes because they have a high number of connections, which is known as degree [25]. Hub genes are identified using the Cytohubba plugin in Cytoscape [24]. From the studies retrieved from the literature review, a total of 87 hub genes were reported, namely CCNB1, RRM2, TOP2A, CDK1, AURKA, BUB1B, NUSAP1, PPARG, KIF11, CCNA2, CCNB2, BIRC5, MELK, DTL, PRC1, TPX2, KIF20A, MAD2L1, CENPF, CEP55, LEP, FGF2, PBK, KIF2C, ASPM, EZH2, CDC20, BUB1, CDCA8, HMMR, KIF23, UBE2C, RACGAP1, DLGAP5, IGF1, PTEN, FOXO1, IK3CA, CDH1, RPS9, RPL11, RPS14, RPL10A, RRM2B, CDC20B, MCM4, NEK2, FABP4, PLIN1, ADIPOQ, CD36, LIPE, LPL, CIDEA, CIDEC, NDC80, CD274, KRT5, KRT16, TP63, KRT7, CXCL5, HGF, TNFRSF8, TNFRSF1B, MMP13, CTSL, CXCL1, CASP1, MAP1B, SQSTM1, IRAK1, TLR3, PYCARD, CASP6, CLEC7A, CENPA, PCNA, KIF4A, KPNA2, TYMS, RRM1, ZWINT, NCAPG, PLK1, KIAA0101, GINS1. The reported hub genes were further examined to identify therapeutic targets using the Therapeutic Target Database (TTD) [23]. Of the 87

reported genes, 51 genes were found to have known therapeutic targets. The remaining 36 hub genes did not have therapeutic targets and were used for further analysis.

2.4 NUSAP1 gene:

Nucleolar and spindle-associated protein 1 (NUSAP1) is a microtubule-associated protein that plays a crucial role in mitosis. Recent studies have demonstrated that NUSAP1 is overexpressed in various cancers with poor prognosis and higher metastasis, which indicates spread to other body parts, and identified it as a biomarker [27]. Functional enrichment analysis demonstrated that NUSAP1 is enriched in biological processes (BP), including cell division, mitotic cell cycle, chromosome separation, nuclear division, regulation of cell cycle, and cell division. NUSAP1 has been reported to play a complicated and sensitized role in cell division and mitotic progression, spindle formation, and stability controlled by phosphorylation. Several studies have demonstrated that NUSAP1 has an important regulatory role in tumors, including cervical cancer [28], colorectal cancer [29], lung cancer [30], prostate cancer [31], and BRCA [32], with high expression of NUSAP1 significantly associated with poor prognosis in BRCA. Functional cell experiments have confirmed that NUSAP1 is linked to the proliferation, migration, and invasion of BRCA cells. Patients with high expression of the NUSAP1 gene are associated with shorter survival. Expression of the NUSAP1 gene varies in different types of BRCA, which concludes that NUSAP1 plays a role in cancer growth and progression and has potential synergistic roles in cellular functions and disease mechanisms [33].

Correlation analysis of gene expression in BRCA. DTL, RRM2, and AURKA are all positively correlated with NUSAP1 (UALCAN). The proliferation, migration, and invasion of gastric cancer cells by inhibiting the mTORC1 signaling pathway [34]. It demonstrated that silencing NUSAP1 induces apoptosis and inhibits the cell cycle in ovarian cancer cells [35]. Other studies have confirmed that NUSAP1 is upregulated in various cancer tissues, contributing to cancer development and progression by promoting epithelial to mesenchymal transition, regulating the transition of the mitotic cell cycle, and modulating multiple immune-related pathways [33]. NUSAP1 is identified as a high expression associated with shorter survival, which suggests a role as a biomarker in the progression of tumor tissues in various cancers [36], and also NUSAP1 plays a role in modulating the tumor microenvironment. NUSAP1 expression was positively correlated with the infiltration of Th2, MDSC, and TFH,

while showing a negative correlation with central and effector memory CD4+ T and NK cells [37]. NUSAP1 appears to be a potential prognostic biomarker and therapeutic target for pan-cancer [38].

CHAPTER 3: MATERIALS & METHODS

3.1 Hub gene data collection from literature:

A literature review was performed to search for previously identified hub genes for breast cancer. A dataset of reported hub genes was created to identify the frequency of each hub gene. The dataset was then filtered to remove the duplicates, and unique hub genes were identified.

3.2 Identification of reported drugs for hub genes using Therapeutic Target Database:

TTD stands for Therapeutic Target Database, which is a database that provides information about the known targets and explored therapeutic protein and nucleic acid targets. It contains information on target-disease associations and drugs in clinical trial stages associated with the targets.

In the present study, the drug target type and drug names were identified for the collected hub genes using the Therapeutic Target Database(https://ttd.idrblab.cn/)[23]. Each hub gene is queried in the search bar for identification of the target. Hub genes that lack details of drugs or the type of drug target were listed for further analysis.

3.3 Construction of protein-protein interaction network using STRING:

STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) [39] is an online database for analyzing and visualizing the protein–protein interactions (PPI). The interacting proteins are plotted as a protein interaction network with a corresponding confidence score.

Hub genes for which no drug target type and drug names have been reported till now are identified. Firstly, the hub genes with no drug target type and drug names were entered into STRING to construct a network for the identification of direct and indirect interactions between them using STRING(https://string-db.org/).

3.4 Identification of hub genes using Cytoscape:

Cytoscape [24] is an open-source software application for visualizing, navigating, and analyzing complex networks that consist of proteins, genes, and their interactions with each other. The Cytohubba plugin is used for the identification of hub genes using various centrality algorithms, such as Degree, Closeness, Betweenness, and Maximal Clique Centrality (MCC), to measure how strongly a node is connected. Genes with high centrality scores are referred to as hub genes.

The protein–protein interaction (PPI) network data was downloaded from the STRING database as a .tsv file. The file was imported into Cytoscape to construct the PPI network and to identify hub genes using the CytoHubba plugin. Maximal Clique Centrality (MCC) algorithms were used to evaluate the connectivity and importance of the genes in the network. The genes with the highest-ranking scores were considered the key hub genes.

3.5 Selection of Candidate Gene based on Breast Cancer and Pan-Cancer Expression Analysis using GEPIA2:

GEPIA2[40] is an online web application for RNA sequencing expression analysis for human cancers and normal tissues from the TCGA and GTEx projects. The normalized and generalized gene expression data are available for the comparison of gene expression between tumor and non-tumor samples.

The Pan-cancer expression level of hub genes from the previous analysis was analyzed by using the GEPIA2 online tool(http://gepia2.cancer-pku.cn/#index). The list of hub genes obtained from Cytohubba was entered in GEPIA2 to analyze the expression of selected hub genes in all cancers. Only tumor data were analyzed in this section to compare gene expression across breast cancers of hub genes, without considering gene expression in normal tissue. Overexpressed genes in breast cancer and pan-cancer were identified using GEPIA2.

3.6 Functional analysis of nsSNPs:

3.6.1 Collection of variants using UniprotKB:

The UniProt Knowledgebase (UniProtKB) is a growing list of manually annotated and reviewed protein sequences (Swiss-Prot) and also contains variants for specific proteins.

UniProtKB [41] was used to retrieve the information on the NUSAP1 gene. The gene name "NUSAP1" was entered in the search bar, and the results were filtered by selecting the organism Homo sapiens to ensure that only human proteins were considered. The non-synonyms single nucleotide polymorphism variants of the NUSAP1 gene were collected for further analysis.

3.6.2 Identification of the most deleterious nsSNPs using Ensembl Variant Effect Predictor:

Ensembl Variant Effect Predictor(https://www.ensembl.org/vep)[42] is a bioinformatics tool used to annotate genomic variants and predict their effect. Variant Effect Predictor provides detailed information about the consequences of each variant, such as missense, nonsense, or synonymous mutations, the affected genes, and transcripts.

SIFT and PolyPhen2 prediction tools predict whether an amino acid substitution is tolerated or deleterious based on sequence homology. The MutationTaster prediction tool evaluate the variant is disease-causing or benign. PROVEAN predicts the functional effect of protein sequence variations, MutationAssessor estimates the impact of amino acid substitutions based on evolutionary conservation, MetaSVM uses machine learning to classify variants as damaging or tolerated, and Clinpred_pred predicts whether the variant is damaging or tolerated. A list of variant identifiers from databases like Uniprot can be submitted to the tool.

3.6.3 Analysis of the effects of nsSNPs on protein stability:

I-Mutant 2.0 (https://folding.biofold.org/cgi-bin/i-mutant2.0.cgi)[43] and MUpro (https://mupro.proteomics.ics.uci.edu)[44] were used to evaluate how non-synonymous SNPs(nsSNPs) impact the structure and stability of the NUSAP1 gene protein structures computationally.

I-Mutant 2.0 is a widely used bioinformatics tool for predicting the effect of single-point mutations on protein stability by using a machine learning approach (Support Vector Machines), and MUpro is a computational tool used to re-evaluate the I-Mutant 2.0 results, as it utilizes both machine learning techniques (i.e., Support Vector Machines and Neural Networks) to classify whether the mutation is stabilizing or destabilizing. I-Mutant 2.0 and MUpro predict the change in Gibbs free energy, which results in the mutation stabilizing (positive ΔΔG values) or destabilizing (negative ΔΔG values) a protein. In I-Mutant 2.0 and MUpro, the protein sequence of the NUSAP1 protein was submitted, the position of the mutations, the replaced amino acids, and the temperature (25°C) and the pH (7) under the physiological conditions.

3.6.4 Analysis of structural effects of nsSNPs using HOPE:

HOPE (https://www3.cmbi.umcn.nl/hope/)[46] is built for automated mutant analysis, providing structural effects of a mutation. It was used to analyze the structural and functional effects of nsSNPs. It evaluates how amino acid substitutions affect the protein stability, size, charge, hydrophobicity, and interactions.

To analyze the structural effects of nsSNPs for the mutated NUSAP1 protein, a target amino acid sequence of NUSAP1 is first provided, then selecting a residue to mutate, selecting a mutation, and selecting a modeling method is applied to generate and compare the three-dimensional structures of wildtype and mutated.

3.6.5 Analysis of NUSAP1 protein evolutionary conservation using ET:

ET stands for Evolutionary Trace Database [45] (https://evolution.lichtargelab.org/), which is a bioinformatics tool that helps scientists to understand how proteins evolve and which part of the protein is important. It ranks amino acid residues of a protein sequence according to their evolutionary conservation.

By providing the protein sequence as input for the evolutionary trace database with default settings, to identify the conserved amino acid residues of a protein sequence retrieved from Uniprot of the NUSAP1 protein for wildtype, T114N variant, and R434M variant.

3.7 Structural analysis of nsSNPs:

3.7.1 Structural modelling of NUSAP1 and its variant structures using AlphaFold:

AlphaFold [47] is used for protein structure modelling, which predicts the three-dimensional(3D) structure of a protein directly from its amino acid sequence. Bioinformatics has been transformed by deep learning models that can accurately predict protein structures comparable to experimental determination for many proteins.

To predict the NUSAP1 wildtype structure and variant structure using the AlphaFold server(https://alphafoldserver.com/). The amino acid sequence of the NUSAP1 protein was provided in FASTA format as input to the AlphaFold server, and it automatically performed multiple sequence alignment, structural prediction, and model refinement using deep learning algorithms. The predicted 3D-structure was generated and downloaded in PDB format and used for further analysis.

3.7.2 Structural Refinement of NUSAP1 and its variant structures using GalaxyRefine:

GalaxyRefine[48] is a widely used web server-based protein structure refinement tool created to improve the quality of predicted 3D protein structures. It helps the tool to improve local geometry, remove steric clashes, and to improve overall structural quality. To refine the predicted structures generated from the AlphaFold using the GalaxyRefine server (https://galaxy.seoklab.org/cgi-bin/submit.cgi?type=REFINE) of NUSAP1 wildtype and its variant structures. Multiple refined models were generated, and the best model was selected based on quality parameters such as RMSD, Clash score, and MolProbity score. The final refined best structure for NUSAP1 wildtype and its mutated structures was used for further analysis.

3.7.3 Structural validation of NUSAP1 and its variant structures using SAVES:

The SAVES (Structure Analysis and Verification Server) [49] is an online resource that assesses the quality and accuracy of structural models of proteins. SAVES is providing a suite of analytical tools, including PROCHECK and ERRAT. PROCHECK evaluates stereochemical quality by analyzing the backbone dihedral angles and generates Ramachandran plots for the backbone and all residues in the model. ERRAT calculates

non-bonded atom–atom interactions throughout the model and provides an overall quality score for the model accuracy.

The refined models of NUSAP1 wild-type and variant structures were obtained from GalaxyRefine and validated on the SAVES(https://saves.mbi.ucla.edu/) online platform. These included checks on the Ramachandran plot using PROCHECK, and overall accuracy using ERRAT to check for non-bonded interactions.

3.8 Analyzing the stability and flexibility using iMODS:

iMODS [52] stands for Internal Coordinates Normal Mode Analysis Server. iMODS is an online computational database and analysis server for studying the structural dynamics of biomacromolecules by performing Normal Mode Analysis (NMA). Thousands of biological macromolecules and complexes can be efficiently analyzed using iMODS for studying protein flexibility and stability, and for evaluating global and collective motion, without extensive computational resources.

The refined structures of wildtype, variant T114N, and variant R434M were analyzed on the iMODS server. (https://imods.iqf.csic.es/) for dynamic properties. The refined structure was uploaded as a structure in PDB format. Normal Mode Analysis (NMA) was performed to identify flexible and rigid domains within the protein–protein complex, and to calculate deformability.

3.9 Prediction of binding pockets using PrankWeb:

PrankWeb [53] is a web-based application that allows to predict and visualization of protein-ligand binding sites. It also allows comparison of the locations of predicted pockets with highly conserved areas and actual binding sites. It uses machine learning algorithms to identify pockets on the protein surface where a ligand can bind. To identify the active binding site of the NUSAP1 protein before performing molecular docking using PrankWeb (https://prankweb.cz/). The prepared protein in .pdb format was uploaded as input to identify pockets on the NUSAP1 protein.

3.10 Virtual Screening, Molecular Docking, and Protein–Ligand Interaction Analysis:

3.10.1 Mcule library screening:

Mcule (https://mcule.com) [55] is used for identifying potential chemical compounds for lead development by structure- and ligand-based virtual screening for designing screening libraries. In modern computer-aided drug discovery (CADD), millions of chemical compounds can be virtually tested for interactions with a protein of interest before laboratory tests are performed. It was used to retrieve chemical compounds for virtual screening against the NUSAP1 protein. The 3D structure of NUSAP1 was obtained from AlphaFold, and after the removal of the water molecules, the heteroatoms, and co-crystallized ligands, it was minimized by adding the hydrogen atoms and performing an energy minimization. Since no ligands were available for the NUSAP1 protein, structure-based virtual screening was performed by using the compound library available in the Mcule Platform.

3.10.2 IMPPAT 2.0 screening:

IMPPAT [55] stands for Indian Medicinal Plants, Phytochemistry, and Therapeutics 2.0 (IMPPAT 2.0) (https://cb.imsc.res.in/imppat/), a manually curated database constructed by digitizing information from more than 100 books on traditional Indian medicine and 7,000+ published research articles. It contains information about Indian medicinal plants, their chemical compounds, and their therapeutic uses. The phytochemical compounds (~17000 compounds) of Indian medicinal plants were retrieved for virtual screening and drug discovery analysis.

3.10.3 Lipinski's Rule of Five-based filtering of compounds:

The compounds were filtered by Lipinski's Rule of Five, which is used for the evaluation of the oral drug likeness of small molecules. Compounds with molecular weight $\leq$ 500 g/mol, LogP $\leq$ 5, hydrogen bond donors $\leq$ 5, and hydrogen bond acceptors $\leq$ 10 are considered good oral drug candidates to select the compounds that were suitable for the docking studies.

3.10.4 Molecular Docking Using AutoDock Vina:

The filtered chemical compounds performed a molecular docking analysis with NUSAP1 protein using AutoDock Vina to predict protein-ligand binding affinities. Docking was performed to evaluate how strongly the NUSAP1 protein binds to the chemical compound. Based on the lowest binding energies, the top 100 phytochemical compounds were selected as lead compounds.

3.10.5 ADME analysis using SwissADME:

The lead compounds were selected based on docking scores and binding energies, and the molecular interactions, including hydrogen bonding and hydrophobic interactions, were analyzed using SwissADME [56] to evaluate drug-likeness and ADME (Absorption, Distribution, Metabolism, and Excretion) properties. Those with good ADME profiles were selected for toxicity testing.

3.10.5 Toxicity Prediction using ProTox 3.0:

After SwissADME filtering, the selected lead compounds were used for predicting toxicity in organs, oral using Protox 3.0 [57]. Based on toxicity prediction, lead compounds showing lower toxicity were identified as promising candidate compounds.

CHAPTER 4: RESULTS & DISCUSSIONS

4.1 Reported hub genes from literature:

Commonly reported hub genes were collected from research papers. The identified hub genes were compiled, and the occurrence of each hub gene was reported. The frequency of the hub genes in the breast cancer datasets was shown in Fig. 4.1.1

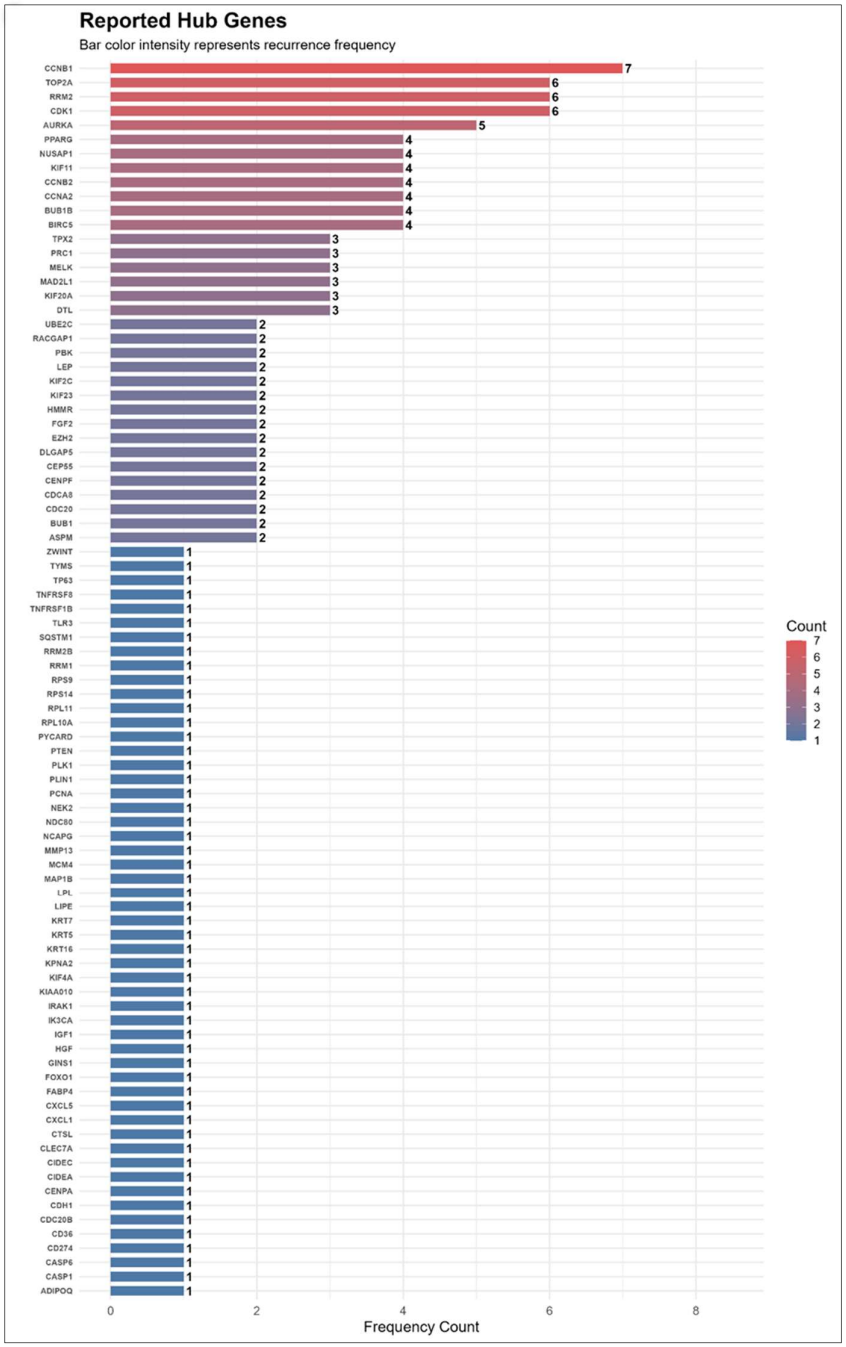


Fig 4.1.1 Horizontal bar plot of the frequency of hub genes and gene names.

Legend: The bar plot provides the frequency of reported hub genes from various studies. The x-axis shows the frequency count (number of studies in which a given gene was reported as a hub gene), and the y-axis shows the gene names. The intensity of the bar's color represents the frequency of gene recurrence, with darker red indicating higher recurrence and lighter blue indicating lower recurrence.

Among the 87 reported hub genes from literature, the highest frequency of 7 for the gene CCNB1 supported its highest degree of connectivity and recurrence. TOP2A, RRM2, and CDK1 have frequencies of 6, indicating that these genes are highly connected in breast cancer-related networks. The AURKA gene has a frequency of 5, implying that it might play an essential role in cancer biology. The genes PPARG, NUSAP1, KIF11, CCNB2, CCNA2, BUB1B, and BIRC5 have a frequency of 4, showing that they are frequently identified as hub genes in different breast cancer datasets.

4.2.1 Reported drugs for hub genes:

Therapeutic drug targets for the reported 87 hub genes were identified using Therapeutic Target Database (TTD). The reported hub genes are segregated based on the presence and absence of therapeutic drugs.

Table 4.2.1.1 List of target types of hub genes and their representative drugs

Sr	Hub gene	Target Name	Target type	Drug
1	CCNB1	G2/mitotic-specific cyclin B1 (CCNB1)	Patented-recorded target	Kenpaullone
2	RRM2	Ribonucleoside-diphosphate reductase M2 (RRM2)	Successful target	Hydroxyurea
3	TOP2A	DNA topoisomerase II (TOP2)	Successful target	Aclarubicin
4	CDK1	Cyclin-dependent kinase 4 (CDK4)	Successful target	Palbociclib
5	AURKA	Aurora kinase A (AURKA)	Clinical trial target	Rosiglitazone + metformin

6	PPARG	Peroxisome proliferator-activated receptor gamma (PPAR-gamma)	Successful target	Pioglitazone
7	KIF11	Kinesin-like protein KIF11 (KIF11)	Clinical trial target	AZD-4877
8	CCNA2	Cyclin A2 (CCNA2)	Literature-reported target	PD-0183812
9	BIRC5	Apoptosis inhibitor survivin (BIRC5)	Clinical trial target	ISIS 23722
10	MELK	Maternal embryonic leucine zipper kinase (MELK)	Clinical trial target	OTSSP167
11	DTL	DDB1- and CUL4-associated factor 2 (DTL)	Literature-reported target	-
12	PRC1	Metabotropic glutamate receptor 3	Clinical trial target	LY404039
13	TPX2	Hepatocellular carcinoma-associated antigen 90 (TPX2)	Literature-reported target	-
14	LEP	Leptin (LEP)	Literature-reported target	-
15	FGF2	Fibroblast growth factor-2 (FGF2)	Successful target	Sucralfate
16	PBK	PDZ binding kinase (PBK)	Literature-reported target	OTS964
17	EZH2	Enhancer of zeste homolog 2 (EZH2)	Successful target	Tazemetostat
18	CDC20	Cell division cycle protein 20 homolog (CDC20)	Preclinical target	Tosyl-l-arginine methyl ester

19	BUB1	BUB1 mitotic checkpoint serine/threonine kinase (BUB1)	Patented-recorded target	PMID 24900824C2-1-1
20	CDCA8	Borealin (CDCA8)	Literature-reported target	-
21	IGF1	Insulin-like growth factor-I (IGF1)	Clinical trial target	Xentuzumab
22	PTEN	Phosphatase and tensin homolog (PTEN)	Literature-reported target	-
23	FOXO1	Forkhead box O1A messenger RNA (FOXO1 mRNA)	Literature-reported target	ISIS 188755
24	IK3CA	PI3-kinase alpha (PIK3CA)	Successful target	Alpelisib
25	CDH1	Epithelial cadherin (CDH1)	Literature-reported target	-
26	NEK2	NIMA-related kinase 2 (NEK2)	Literature-reported target	GSK579289A
27	FABP4	Fatty acid-binding protein 4 (FABP4)	Patented-recorded target	2-(4-Fluoro -indole-1 -sulfonyl) -benzoic acid
28	PLIN1	Lipid droplet-associated protein (PLIN1)	Literature-reported target	-
29	ADIPOQ	Adiponectin (ADIPOQ)	Literature-reported target	-
30	CD36	Platelet glycoprotein 4 (CD36)	Literature-reported target	-
31	LIPE	Hormone-sensitive lipase (LIPE)	Successful target	Benzyl Benzoate

32	LPL	Lipoprotein lipase (LPL)	Successful target	Clofibrate
33	CD274	Programmed cell death 1 ligand 1 (PD-L1)	Successful target	Bavencio
34	CXCL5	CXCL5/CXCR2 axis	Literature-reported target	-
35	HGF	Hepatocyte growth factor (HGF)	Clinical trial target	VM-202
36	TNFRSF8	Lymphocyte activation antigen CD30 (TNFRSF8)	Successful target	Brentuximab vedotin
37	TNFRSF1B	Tumor necrosis factor receptor type II (TNF-R2)	Literature-reported target	-
38	MMP13	Matrix metalloproteinase	Clinical trial target	Curcumin
39	CTSL	Cathepsin L (CTSL)	Clinical trial target	KGP94
40	CXCL1	C-X-C motif chemokine 1 (CXCL1)	Literature-reported target	-
41	CASP1	Caspase-1 (CASP1)	Clinical trial target	Belnacasan
42	MAP1B	Microtubule-associated protein (MAP)	Successful target	Demecolcine
43	SQSTM1	Sequestosome-1 p62 (SQSTM1)	Literature-reported target	IMX-942
44	IRAK1	IL-1 receptor-associated kinase 1 (IRAK1)	Patented-recorded target	IRAK-1/4 inhibitor

45	TLR3	HUMAN toll-like receptor 3 (TLR3)	Clinical trial target	Rintatolimod
46	CASP6	Caspase 6 messenger RNA (CASP6 mRNA)	Literature-reported target	ISIS 134499
47	PCNA	Proliferating cell nuclear antigen (PCNA)	Clinical trial target	ATX-101
48	TYMS	Thymidylate synthase (TYMS)	Clinical trial target	Plevitrexed (R)-isomer
49	RRM1	RRM1 messenger RNA (RRM1 mRNA)	Discontinued target	LOR-2501
50	ZWINT	ZW10 interactor (ZWINT)	Literature-reported target	-
51	PLK1	Polo-like kinase 1 (PLK1)	Clinical trial target	Volasertib

The table represents the hub genes that have drug target type and drug names. A total of 87 hub genes were reported from the literature, out of which 51 have drug target type or drug names reported in TTD.

Table 4.2.1.2 List of hub genes that lack target types and their representative drugs

Hub genes	GINS1, KIAA010, NCAPG, KIF4A, KPNA2, CLEC7A, CENPA, PYCARD, KRT5, KRT1, TP63, KRT7, CIDEA, CIDEC, NDC80, RPS9, RPL11, RPS14, RPL10A, RRM2B, CDC20B, MCM4, HMMR, KIF23, UBE2C, RACGAP1, DLGAP5, KIF2C, ASPM, KIF20A, MAD2L1, CENPF, CEP55, BUB1B, CCNB2, NUSAP1

The table represents the hub genes that lack drug target type and drug names. A total of 87 hub genes were reported from the literature, and for 36 hub genes lack of drug target type and drug names in TTD.

4.3 PPI network of selected hubs:

A protein-protein interaction network was constructed for reported hub genes that lack therapeutic drug target type and drug data using STRING, shown in Figure 4.3.1.

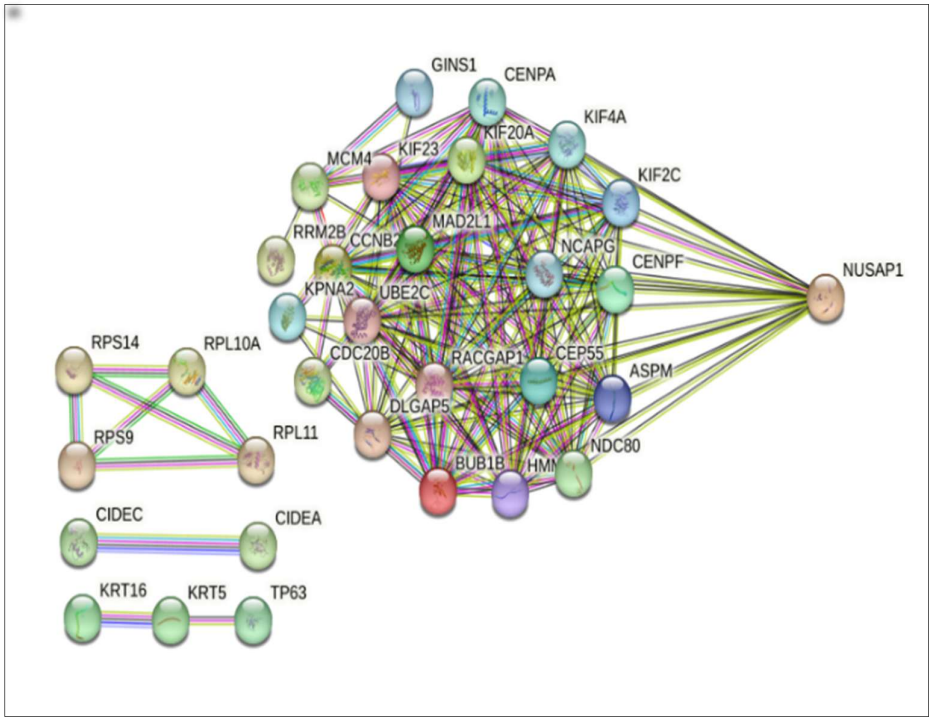


Fig 4.3.1 Protein-protein interaction network.

Legend: In the protein-protein interaction network, each node represents a gene, and the edges between nodes represent the various types of interactions for the genes. The circular layout is used for displaying the relationships between genes in the network.

The protein interaction network shows that most of the proteins are closely connected to each other. The edges with higher combined STRING scores are shown as thicker and darker links, while lower-confidence associations are shown as thinner and lighter links. A large number of genes are involved in these interactions; the genes likely interact with one another in the regulation of common biological processes. Proteins that work together in the cell frequently appear as connected groups in the network.

4.4 Ranking of selected hub genes:

Hub genes were identified using the Cytohubba plugin of Cytoscape for the constructed gene-gene interaction network shown in Fig. 4.4.1.

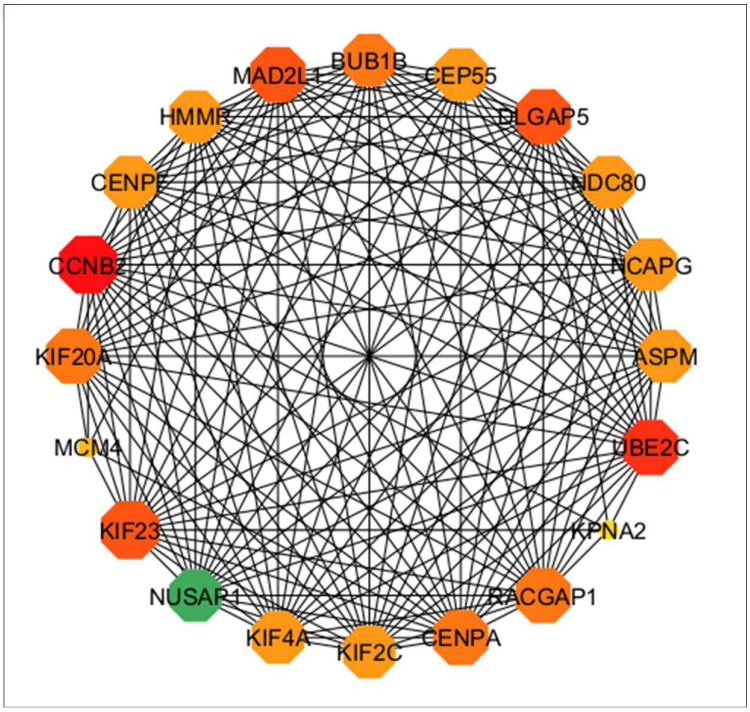


Fig 4.4.1 Hub genes identified using Cytoscape.

The colored nodes are genes, and the edges represent interactions between the nodes. The red color nodes have the highest MCC scores, which means that these genes are the top-ranking hub genes having the largest number of interactions with other genes. The orange color nodes represent genes with moderately high MCC scores, meaning they have slightly fewer interactions than top-ranking hub genes, while the yellow color nodes represent genes with relatively lower MCC scores among the selected hubs.

The NUSAP1 gene appears in green color, which means to highlight the gene from the other hub genes in the network. The size of the nodes indicates the interactions between genes; the bigger the size of the node has more interactions whereas the smaller the node, has lesser interactions. The network generated from a set of genes involved in mitosis revealed that CCNB2, BUB1B, UBE2C, LGAP5, MAD2L1, and KIF23 genes received a large number of connections within the network. These genes are required for the proliferation of cells. The NUSAP1 gene was found to be associated with several mitotic genes, including KIF23, KIF4A, RACGAP1, CENPA, and KIF2C

genes, which are involved in spindle formation and chromosome separation during cell division.

4.5.1 Breast Cancer Expression Analysis of Top-Ranked Hub Genes:

Expression of identified hub genes in breast cancer was analyzed using GEPIA2. The heatmap (Fig. 4.6.1) represents the top 10 ranking hub genes expression in breast cancer.

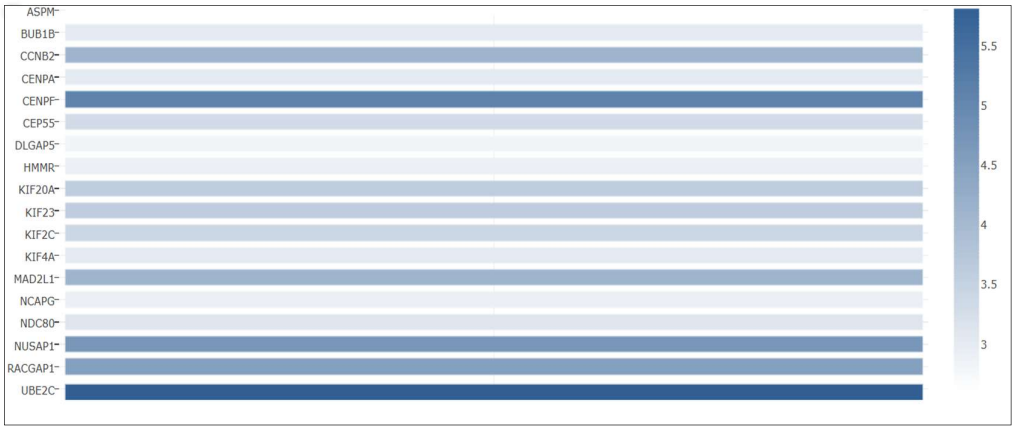


Fig 4.5.1.1 Top-ranked hub genes expression in Breast Cancer using GEPIA

Legend: The rows represent hub genes, and the columns represent breast cancer. Each cell represents the expression of the hub gene in a specific cancer type. The darker the shade, the higher the expression of a particular gene in breast cancer. The lighter the shade, the lower the expression of a gene in breast cancer. The scale on the right ranges from 0 to 7, indicating levels of gene expression on a log2(TPM+1) scale.

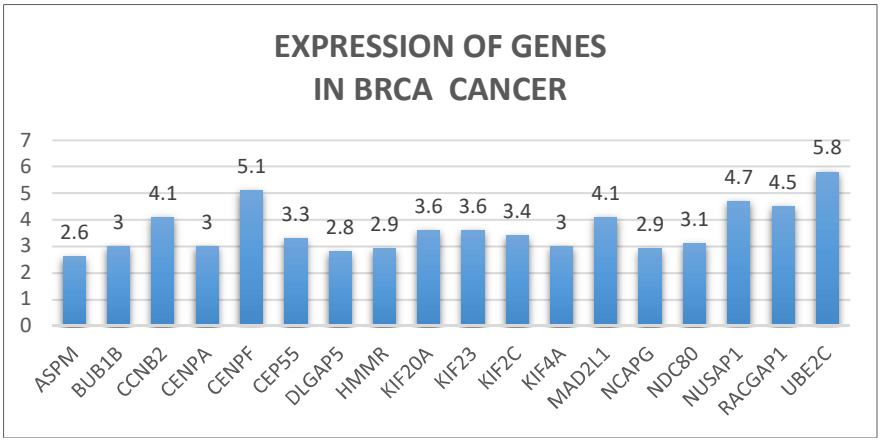


Fig 4.5.1.2 Expression of Genes in Breast Cancer using GEPIA2

Legend: The bar graph (Fig. 4.5.1.2) represents the expression of the top 10 ranking hub genes, which were retrieved from GEPIA2. The X-axis shows the genes, and the Y–axis represents the expression of each gene in breast cancer.

The bar plot shows that UBE2C has higher expression in breast cancer, CENPF, and NUSAP1 have slightly less expression as compared to UBE2C.

4.5.2 Pan-cancer expression analysis of top-ranking hub genes:

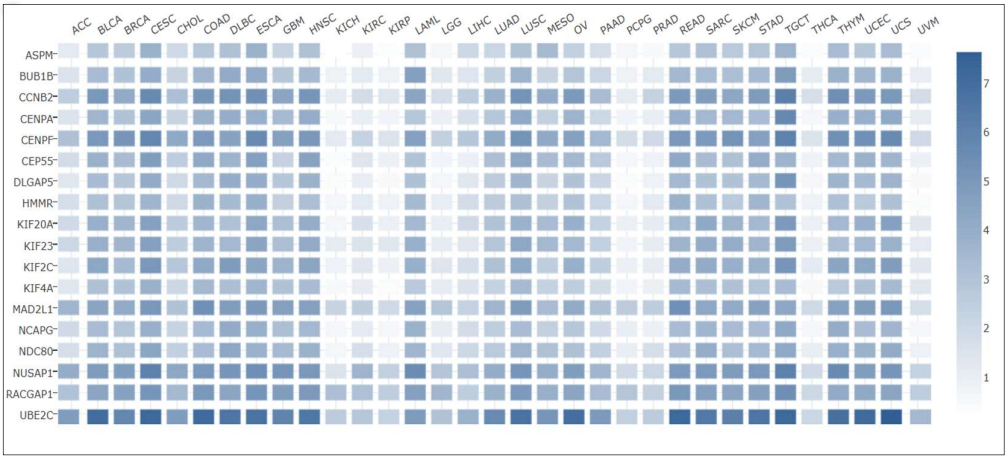


Fig 4.5.2.1 Pan-cancer expression of hub genes using GEPIA

Legend - The rows represent hub genes, and the columns represent types of cancer. Each cell represents the expression of the hub gene in a specific cancer type. The darker the shade, the higher the expression of a particular gene in a specific type of cancer. The lighter the shade, the lower the expression of a particular gene in a specific type of cancer. The scale on the right ranges from 0 to 7, indicating levels of gene expression.

UBE2C is highly expressed in breast cancer, as well as in all 33 cancers; however, previous studies have already reported it as a biomarker in pan-cancer. CENPF is the next highly expressed gene in breast cancer, but not highly expressed in all cancers. NUSAP1 is a highly expressed gene in breast cancer and also in 33 cancers, as compared to others. So, we selected the NUSAP1 gene as a candidate gene for further analysis.

4.6 Functional Analysis of nsSNPs:

4.6.1 Variants of NUSAP1:

UniProt variant of the NUSAP1 gene, which contains 460 variants of NUSAP1 that were collected for further analysis using UniProt. Each entry represents one observed variant of a protein, which is characterized by a Variant ID, which is a unique number for each variant, and the position in the NUSAP1 sequence. It enables users to retrieve disease-associated variants of the NUSAP1 protein.

4.6.2 Identified the most deleterious nsSNPs of the NUSAP1 gene:

Variants collected from UniProt were entered into the Variant Effect Predictor to identify the pathogenic variants. The Ensembl Variant Effect Predictor (VEP)single-nucleotide polymorphisms (SNPs) were submitted to the input field as dbSNP rsIDs. VEP output highlights in silico predictions of the functional impact of variants, using tools such as SIFT, PolyPhen-2, MutationTaster, and other consequence predictors available through Ensembl. 3 variants are likely to be pathogenic from 460 variants identified using prediction tools.

4.6.3 Variant stability analysis:

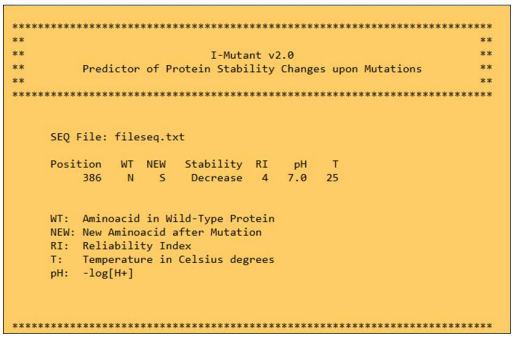


Fig 4.6.3.1 I-Mutant 2.0 prediction of protein stability for N386S variant

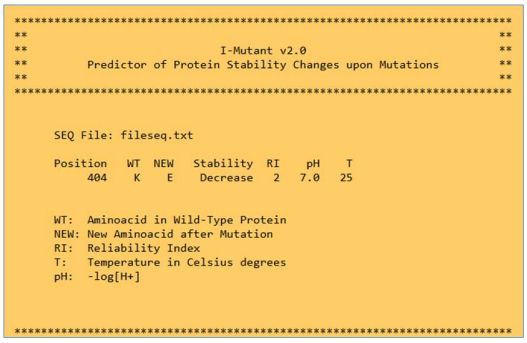


Fig 4.6.3.2 I-Mutant 2.0 prediction of protein stability for K404E variant

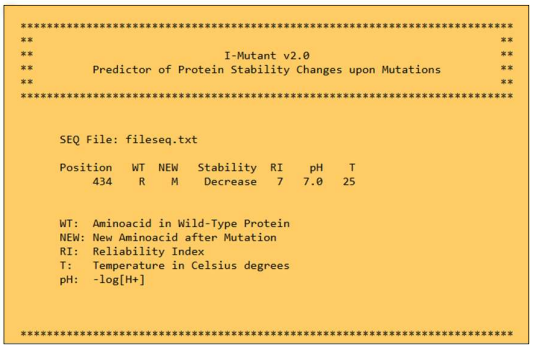


Fig 4.6.3.3 I-Mutant 2.0 prediction of protein stability for R434M variant

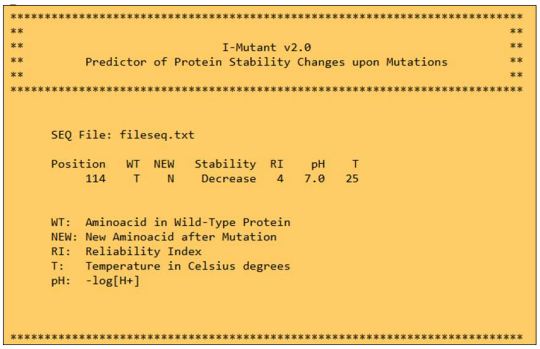


Fig 4.6.3.4 I-Mutant 2.0 prediction of protein stability for T114N variant

Table 4.6.3.1 I-Mutant 2.0 results

I-Mutant 2.0 Results			
Position	**Wildtype residue**	**New residue**	**Stability**
114	T	N	Decrease
386	N	S	Decrease
404	K	E	Decrease
434	R	M	Decrease

Legend: WT means wildtype- position of amino acid residue in the original sequence. NEW means a new amino acid residue that needs to be replaced with a wild-type amino acid residue. Stability- change in Gibbs free energy- decreases the mean destabilizing of the structure and increases the mean stabilizing the structure. Reliability index, confidence level of protein stability changes, which ranges from 0 - 2(lower reliability), 3 – 6(moderate reliability), 7- 10(high reliability). pH - 7 and T – 25 °C are used with default settings as they are standard physiological conditions.

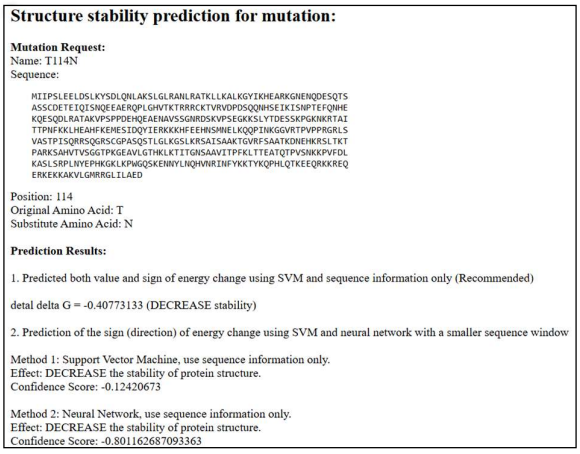
Using I-Mutant 2.0, protein stability predictions for the NUSAP1 protein indicated that the four variations T114N, N386S, K404E, and R434M destabilize the protein

structure.

The protein stability analysis using MUpro for the NUSAP1 variants T114N, N386S, K404E, and R434M.

Table 4.6.3.2 MUpro Results

MUpro Results				
Position	**Wildtype residue**	**New residue**	**Stability**	**ΔΔG value**
114	T	N	Decrease	-0.407
386	N	S	Decrease	-1.332
404	K	E	Decrease	-0.343
434	R	M	Decrease	-0.539



Structure stability prediction for mutation:

Mutation Request:
Name: T114N
Sequence:

MIIPSLEELDSLKYSDLQNLAKSLGLRANLRATKLLKALKGYIKHEARKGNENQDESQTS
ASSCDETEIQISNQEEAERQPLGHVTKTRRRCKTVRVDPDSQQNHSEIKISNPTEFQNHE
KQESQDLRATAKVPSPPDEHQEAENAVSSGNRDSKVPSEGKKSLYTDESSKPGKNKRTAI
TTPNFKKLHEAHFKEMESIDQYIERKKKHFEEHNSMNELKQQPINKGGVRTPVPPRGRLS
VASTPISQRRSQGRSCGPASQSTLGLKGSLKRSAISAAKTGVRFSAATKDNEHKRSLTKT
PARKSAHVTVSGGTPKGEAVLGTHKLKTITGNSAAVITPFKLTTEATQTPVSNKKPVFDL
KASLSRPLNYEPHKGKLKPWGQSKENNYLNQHVNRINFYKKTYKQPHLQTKEEQRKKREQ
ERKEKKAKVLGMRRGLILAED

Position: 114
Original Amino Acid: T
Substitute Amino Acid: N

Prediction Results:

1. Predicted both value and sign of energy change using SVM and sequence information only (Recommended)

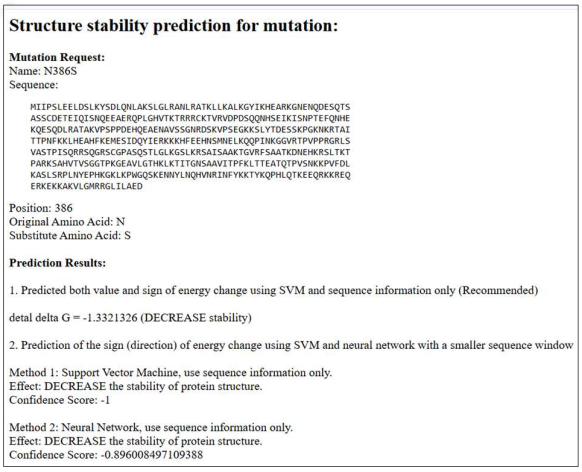
detal delta G = -0.40773133 (DECREASE stability)

2. Prediction of the sign (direction) of energy change using SVM and neural network with a smaller sequence window

Method 1: Support Vector Machine, use sequence information only.
Effect: DECREASE the stability of protein structure.
Confidence Score: -0.12420673

Method 2: Neural Network, use sequence information only.
Effect: DECREASE the stability of protein structure.
Confidence Score: -0.801162687093363

Fig 4.6.3.5 MUpro prediction of protein stability of T114N variant



Structure stability prediction for mutation:

Mutation Request:
Name: N386S
Sequence:

MIIPSLEELDSLKYSDLQNLAKSLGLRANLRATKLLKALKGYIKHEARKGNENQDESQTS
ASSCDETEIQISNQEEAERQPLGHVTKTRRRCKTVRVDPDSQQNHSEIKISNPTEFQNHE
KQESQDLRATAKVPSPPDEHQEAENAVSSGNRDSKVPSEGKKSLYTDESSKPGKNKRTAI
TTPNFKKLHEAHFKEMESIDQYIERKKKHFEEHNSMNELKQQPINKGGVRTPVPPRGRLS
VASTPISQRRSQGRSCGPASQSTLGLKGSLKRSAISAAKTGVRFSAATKDNEHKRSLTKT
PARKSAHVTVSGGTPKGEAVLGTHKLKTITGNSAAVITPFKLTTEATQTPVSNKKPVFDL
KASLSRPLNYEPHKGKLKPWGQSKENNYLNQHVNRINFYKKTYKQPHLQTKEEQRKKREQ
ERKEKKAKVLGMRRGLILAED

Position: 386
Original Amino Acid: N
Substitute Amino Acid: S

Prediction Results:

1. Predicted both value and sign of energy change using SVM and sequence information only (Recommended)

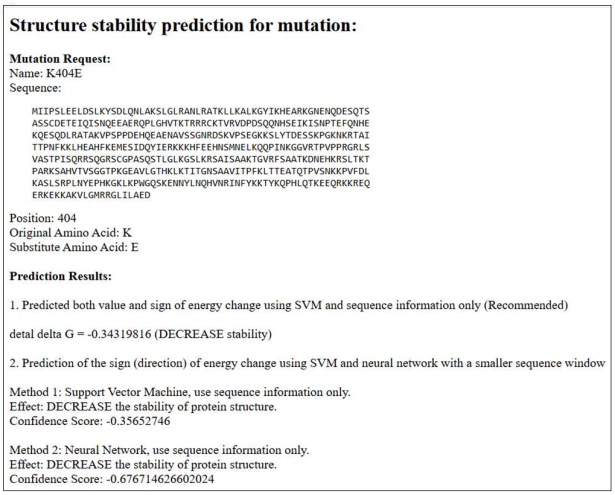
detal delta G = -1.3321326 (DECREASE stability)

2. Prediction of the sign (direction) of energy change using SVM and neural network with a smaller sequence window

Method 1: Support Vector Machine, use sequence information only.
Effect: DECREASE the stability of protein structure.
Confidence Score: -1

Method 2: Neural Network, use sequence information only.
Effect: DECREASE the stability of protein structure.
Confidence Score: -0.896008497109388

Fig 4.6.3.6 MUpro prediction of protein stability of N386S variant



Structure stability prediction for mutation:

Mutation Request:
Name: K404E
Sequence:

```
MIIPSLEELDSLKYSDLQNLAKSLGLRANLRATKLLKALKGYIKHEARKGNENQDESQTS
ASSCDETEIQISNQEEAERQPLGHVTKTRRRCKTVRVDPDSQQNHSEIKISNPTEFQNHE
KQESQDLRATAKVPSPPDEHQEAENAVSSGNRDSKVPSEGKKSLYTDESSKPGKNKRTAI
TTPNFKKLHEAHFKEMESIDQYIERKKKHFEEHNSMNELKQQPINKGGVRTPVPPRGRLS
VASTPISQRRSQGRSCGPASQSTLGLKGSLKRSAISAAKTGVRFSAATKDNEHKRSLTKT
PARKSAHVTVSGGTPKGEAVLGTHKLKTITGNSAAVITPFKLTTEATQTPVSNKKPVFDL
KASLSRPLNYEPHKGKLKPWGQSKENNYLNQHVNRINFYKKTYKQPHLQTKEEQRKKREQ
ERKEKKAKVLGMRRGLILAED
```

Position: 404
Original Amino Acid: K
Substitute Amino Acid: E

Prediction Results:

1. Predicted both value and sign of energy change using SVM and sequence information only (Recommended)

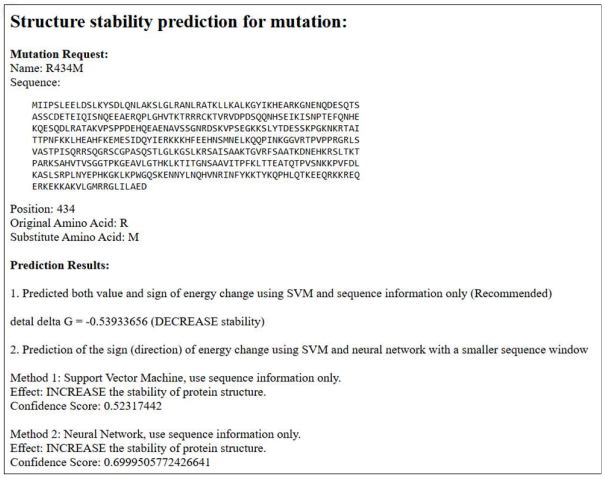
detal delta G = -0.34319816 (DECREASE stability)

2. Prediction of the sign (direction) of energy change using SVM and neural network with a smaller sequence window

Method 1: Support Vector Machine, use sequence information only.
Effect: DECREASE the stability of protein structure.
Confidence Score: -0.35652746

Method 2: Neural Network, use sequence information only.
Effect: DECREASE the stability of protein structure.
Confidence Score: -0.676714626602024

Fig 4.6.3.7 MUpro prediction of protein stability of the K404E variant



Structure stability prediction for mutation:

Mutation Request:
Name: R434M
Sequence:

```
MIIPSLEELDSLKYSDLQNLAKSLGLRANLRATKLLKALKGYIKHEARKGNENQDESQTS
ASSCDETEIQISNQEEAERQPLGHVTKTRRRCKTVRVDPDSQQNHSEIKISNPTEFQNHE
KQESQDLRATAKVPSPPDEHQEAENAVSSGNRDSKVPSEGKKSLYTDESSKPGKNKRTAI
TTPNFKKLHEAHFKEMESIDQYIERKKKHFEEHNSMNELKQQPINKGGVRTPVPPRGRLS
VASTPISQRRSQGRSCGPASQSTLGLKGSLKRSAISAAKTGVRFSAATKDNEHKRSLTKT
PARKSAHVTVSGGTPKGEAVLGTHKLKTITGNSAAVITPFKLTTEATQTPVSNKKPVFDL
KASLSRPLNYEPHKGKLKPWGQSKENNYLNQHVNRINFYKKTYKQPHLQTKEEQRKKREQ
ERKEKKAKVLGMRRGLILAED
```

Position: 434
Original Amino Acid: R
Substitute Amino Acid: M

Prediction Results:

1. Predicted both value and sign of energy change using SVM and sequence information only (Recommended)

detal delta G = -0.53933656 (DECREASE stability)

2. Prediction of the sign (direction) of energy change using SVM and neural network with a smaller sequence window

Method 1: Support Vector Machine, use sequence information only.
Effect: INCREASE the stability of protein structure.
Confidence Score: 0.52317442

Method 2: Neural Network, use sequence information only.
Effect: INCREASE the stability of protein structure.
Confidence Score: 0.6999505772426641

Fig 4.6.3.8 MUpro prediction of protein stability of R434M variant

Legend: ΔΔG energy means the change in Gibbs free energy of protein folding between a wild type and a mutated type. ΔΔG ranges from -5 kcal/mol to +5 kcal/mol. ΔΔG < 0, mutation decreases the stability of the protein, which leads to destabilization. ΔΔG > 0, mutation increases the stability of the protein, which leads to stabilization of the structure. The confidence score is calculated to predict the protein stability upon mutations. Confidence score ranges from -1 to +1. A confidence score > 0 mutation increases the stability of the protein, and a confidence score < 0 mutation decreases the stability of the protein.

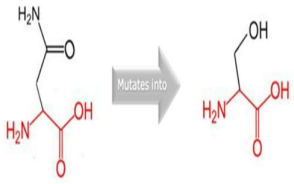
Both I-Mutant 2.0 and MUpro predicted that all four variants of the NUSAP1 protein decrease the protein stability, which indicates that each mutation has a destabilizing effect on the protein. The N386S variant causes the greatest amount of destabilization

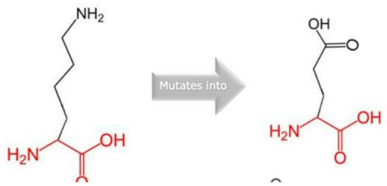
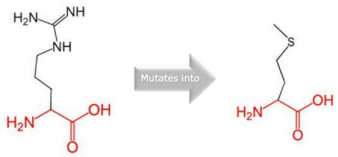
of the four variants, with a ΔΔG of −1.332. The R434M variant was found to cause the second greatest amount of destabilization with a ΔΔG of −0.539, followed by the T114N variant with a ΔΔG of −0.407. The K404E variant was found to cause the least amount of destabilization with a ΔΔG of −0.343. Ranking the variants by the predicted change in folding free energy yields the following order of impact: N386S > R434M > T114N > K404E - from most destabilizing to least destabilizing.

4.6.4 Structural effects analysis of variants:

The MetaRNN score is developed to predict the pathogenicity of variants. It ranges from 0.0 to 1.0; the higher the score, the more likely it is to be pathogenic.

Table 4.6.4.1 HOPE results of Wildtype NUSAP1 and its variants (T114N, N386S, K404E, R434M)

Sr	Variant	Amino acid	MetaRNN score
1	T114N	 Threonine → Asparagine	0.039003207
		- Mutant reside is bigger than the wild type residue. - Wildtype is more hydrophobic than the mutant residue. - Mutation is located in a disordered region. - Wildtype residue is not conserved at this position. - The mutant residue is bigger; this might lead to bumps.	
2	N386S	 Asparagine → Serine	0.3353392
		- Mutant residue is smaller than wild-type residue. - Mutant residue is more hydrophobic than wildtype residue. - Mutant residue is located within a stretch of residues at a special motif- KEN box, a highly conserved position.	

		- Mutant residue is smaller, which might lead to loss of interactions, resulting in loss of hydrogen bonds or disturb correct folding.	
3	K404E	 Lysine → Glutamic acid	0.72709656
		- Mutant residue is smaller than the wild-type residue. - Wild-type residue charge was positive, and mutant residue charge is negative. - Mutation is located in the disordered region. - Mutant residue is located near a highly conserved position. - Mutation introduces a residue with a charge opposite to the wild type, which causes repulsion with other residues in the protein. - Mutant residue is smaller; this might lead to loss of interactions.	
4	R434M	 Arginine → Methionine	0.8230145
		- Mutant residue is smaller than the wild-type residue. - Wild-type residue charge was positive; the mutant residue charge is neutral. - Mutant residue is more hydrophobic than the wild-type residue. - A mutant residue was found at a conserved position; a mutation of 100% conserved residue is usually damaging for a protein.	

		- Mutation introduces a more hydrophobic residue at this position, which results in loss of hydrogen bonds or disturbs correct folding. - The charge of the wild type will be lost; this can cause loss of interactions with other residues.

Based on MetaRNN scores and structural effects predicted by the HOPE server, the four variants can be ranked according to pathogenicity as: R434M > K404E > N386S > T114N. The major changes in charge, hydrophobicity, and conservation status. N386S may moderately affect the function of the protein, since it is within the KEN-box, a conserved structural motif, which could interfere with specific interactions. T114N is likely to have the least effect, since it is in a non-conserved region of the protein that is already disordered. Of the four variants considered, R434M is by far the strongest candidate for being pathogenic, followed by K404E, N386S, and lastly T114N.

4.6.5 Protein evolutionary conservation analysis of wildtype and its variants:

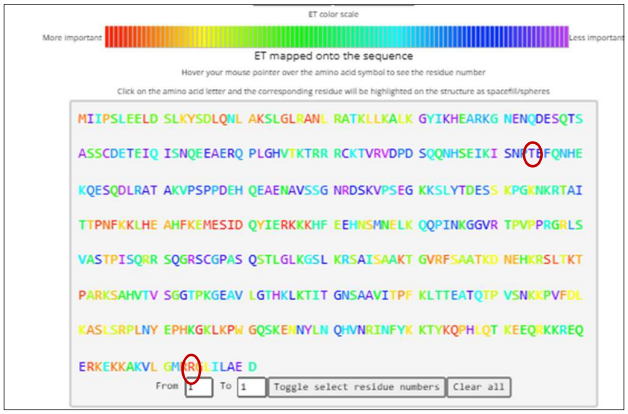


Fig 4.6.4.1 ET results for identification of conserved residues for WT NUSAP1

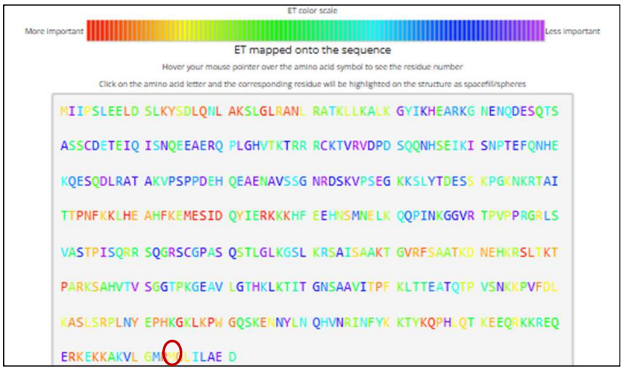


Fig 4.6.4.2 ET results for identification of conserved residues for R434M variant

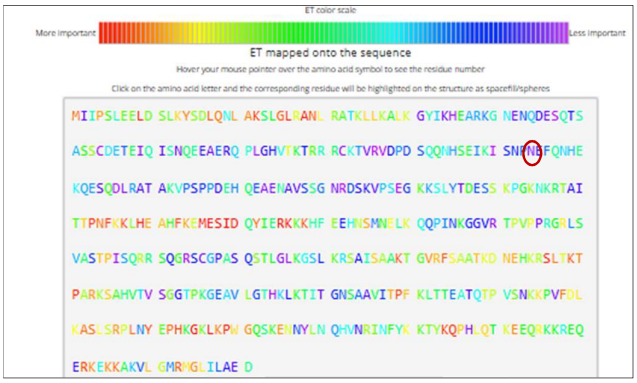


Fig 4.6.4.2 ET results for identification of conserved residues for T114N variant

The wild-type protein has the reddest regions on the evolutionary trace map, indicating it contains the most conserved residues. Mutations at the conserved residues are usually harmful. The R434M variant proteins have changed from the most conserved residue (R434 -red color) to a less conserved residue (434M- yellow), which is harmful as it alters an amino acid, and the T114N variant protein has no changes.

4.7 Structural analysis of nsSNPs:

4.7.1 Modelling of NUSAP1 and its variant structures using AlphaFold:

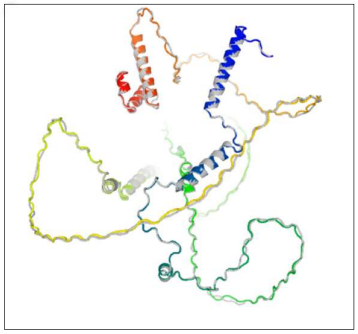


Fig 4.7.2.1 NUSAP1 Wildtype structure

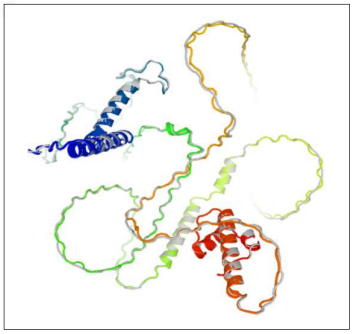


Fig 4.7.2.2 R434M variant structure

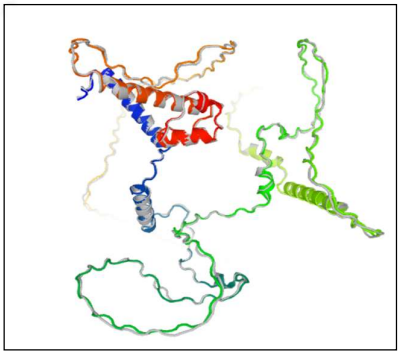


Fig 4.7.2.3 T114N variant structure

By submitting the sequences of NUSAP1 in the structure prediction software AlphaFold, 3D models for wild-type and the T114N and R434M variants were predicted. The wild-type model can serve as a reference model for the protein and be compared to the two variants.

4.7.2 Structural Refinement studies of NUSAP1 and its variant structures:

Using GalaxyRefine to improve the structural quality for the wildtype NUSAP1 and its R434M and T114N variants. For each of these structures, 5 refined structures were generated (Fig. 4.14.1, 4.14.2, 4.14.3) with GDT-HA, RMSD, and MolProbity, clash scores, poor rotamer scores, and Ramachandran favored region percentages based on scores model 1 were downloaded for further analysis.

Structure Information

Model	GDT-HA	RMSD	MolProbity	Clash score	Poor rotamers	Rama favored
Initial	1.0000	0.000	2.456	5.1	2.8	70.2
MODEL 1	0.8532	0.755	0.926	0.8	0.8	97.0
MODEL 2	0.8475	0.765	1.188	1.4	0.0	95.4
MODEL 3	0.8515	0.756	0.919	0.7	0.3	96.8
MODEL 4	0.8532	0.745	0.984	1.0	0.3	96.8
MODEL 5	0.8435	0.805	1.146	1.2	0.0	95.7

Fig 4.7.3.1 NUSAP1 Wildtype structure refinement information

Structure Information

Model	GDT-HA	RMSD	MolProbity	Clash score	Poor rotamers	Rama favored
Initial	1.0000	0.000	2.095	5.1	1.0	73.1
MODEL 1	0.8605	0.729	0.920	1.1	0.3	97.5
MODEL 2	0.8543	0.727	0.818	1.1	0.3	98.2
MODEL 3	0.8605	0.728	0.926	1.0	0.3	97.3
MODEL 4	0.8452	0.759	0.830	1.1	0.0	97.9
MODEL 5	0.8509	0.738	0.932	1.4	0.0	97.7

Fig 4.7.3.2 R434M variant structure refinement information

Structure Information

Model	GDT-HA	RMSD	MolProbity	Clash score	Poor rotamers	Rama favored
Initial	1.0000	0.000	2.280	4.7	1.8	69.7
MODEL 1	0.8345	0.773	1.114	1.4	0.3	96.4
MODEL 2	0.8367	0.793	1.154	1.4	0.8	95.9
MODEL 3	0.8231	0.812	1.014	1.1	0.0	96.8
MODEL 4	0.8203	0.801	1.021	0.8	0.3	96.1
MODEL 5	0.8350	0.776	1.074	0.8	0.3	95.4

Fig 4.7.3.3 T114N variant structure refinement information

Legend: GDT-HA stands for Global Distance Test – High Accuracy score, which measures the accuracy of a refined protein model's carbon alpha atoms, where the higher the value, the more accurate the model quality. The Root Mean Square Deviation score measures the distance between the atoms after superimposing the initial model with the refined model. RMSD score ranges from 0 to 1; the lower the score, the more similar to the initial structure. MolProbity score, which measures the physical correctness and overall quality. MolProbity score ranges from 0 to 4; a lower score indicates a better-quality model. Clash scores measure the number of steric clashes between atoms. Clash scores range from 0 – 10; the lower the score less steric clashes. The Ramachandran favored region percentage is the percentage of residues present in favored regions of the Ramachandran plot of protein structure. The higher the favored % value, the more stable the structure and the better the protein backbone geometry.

After refinement of structures, based on the score of parameters, model 1 is used for further analysis, as it has good quality and accuracy of predicted structure.

4.7.3 Structural validation studies of NUSAP1 and its variants:

The structural quality of the refined protein models was further validated by assessing non-bonded interactions using the ERRAT program in the SAVES v6.1 server.

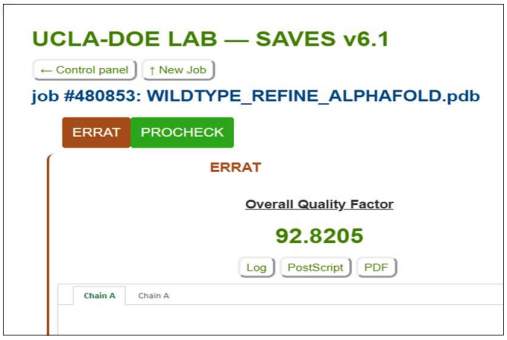


Fig 4.7.4.1 NUSAP1 Wildtype structure validation

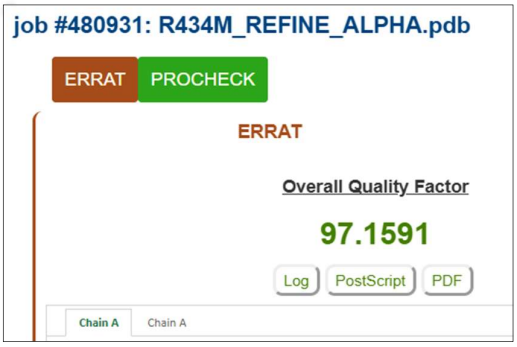


Fig 4.7.4.2 R434M variant structure validation



Fig 4.7.4.3 T114N variant structure validation

The ERRAT scores of all three models were above 90, which are considered to be good models. For the wildtype, the overall quality factor is 92.8205. For T114N, we find an overall quality factor of 94.5652, indicating better non-bonded interactions than the wildtype. The best overall quality factor for all tested variants is 97.1591, achieved for R434M. The Ramachandran plot analysis generated by PROCHECK using the selected MODEL 1 structures provides an assessment of the stereochemical quality for both wild-type and variant proteins.

Fig 4.7.4.4 Wildtype (WT): The wildtype structure contains 95.9% residues in the most favored regions of the Ramachandran map, in addition to 3.3% residues in allowed regions of the map. Only 0.5% residues are in disallowed regions of the map. The WT model is far above the threshold for acceptable quality (>90% favored regions).

Fig 4.7.4.5 R434M variant: For the R434M variant, 95.6% of residues are in the most favored regions of the Ramachandran map, 3.3% in additionally allowed regions, and 0.3% in disallowed regions, representing a slight improvement upon the wildtype with respect to the number of outliers.

Fig 4.7.4.6 T114N variant: The quality of the T114N variant is only marginally decreased compared to WT and R434M, with 93.6% of residues in the most favored regions of the Ramachandran plot, 4.4% in additionally allowed regions, and 0.8% in disallowed regions.

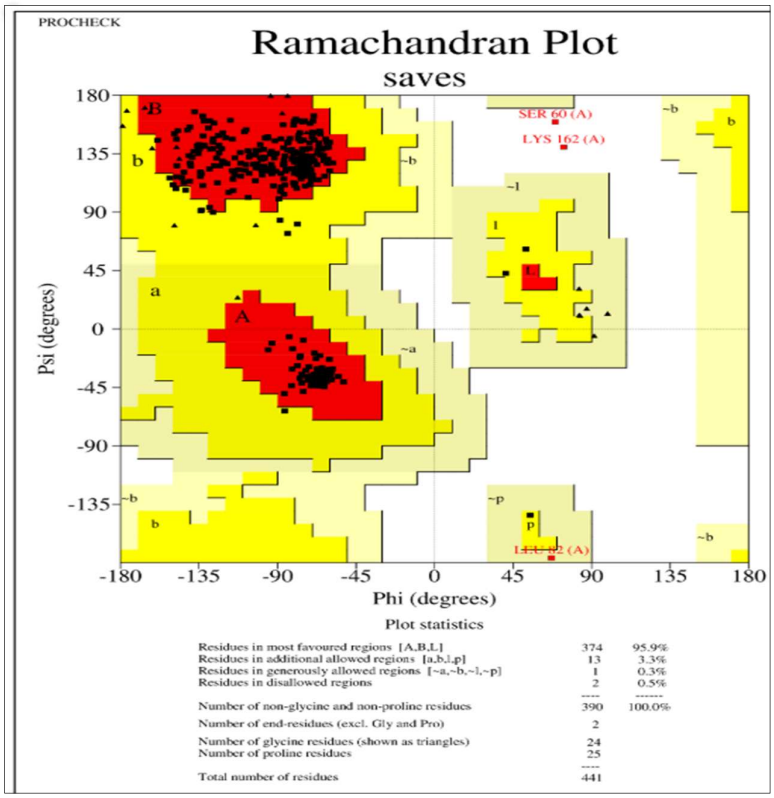


Fig 4.7.4.4 NUSAP1 Wildtype Ramachandran Plot

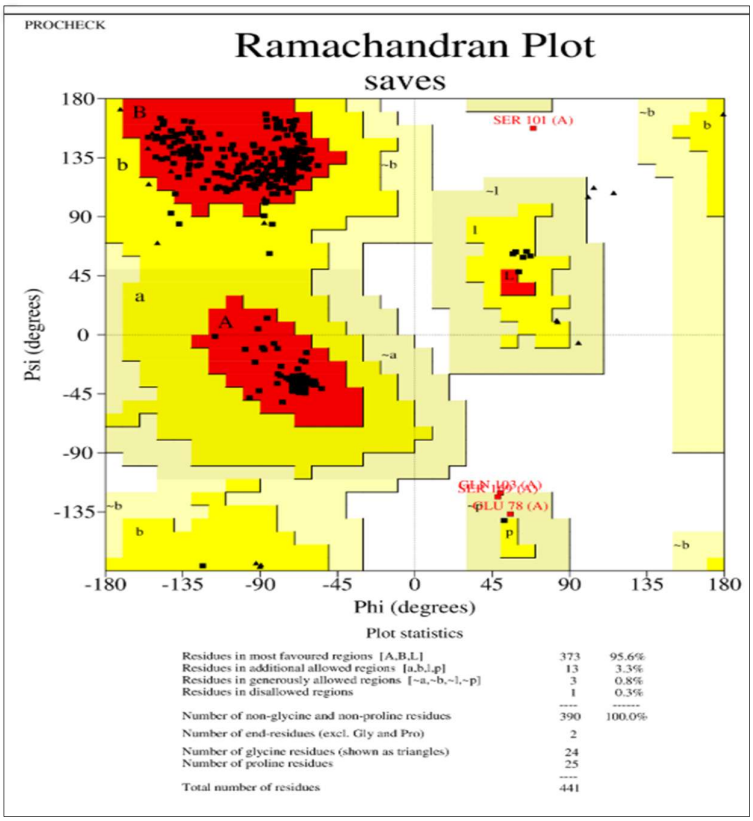


Fig 4.7.4.5 R434M variant Ramachandran Plot

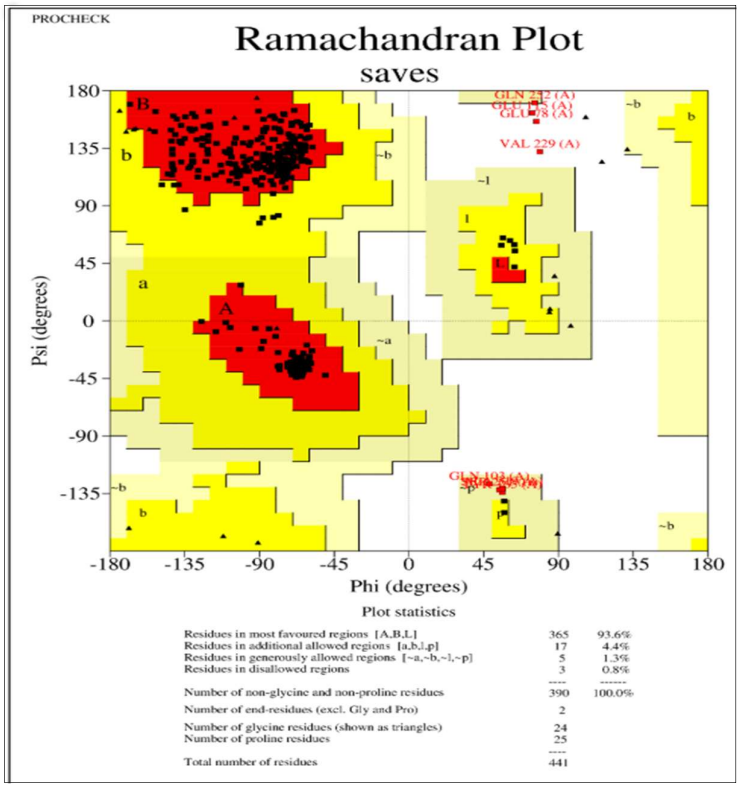


Fig 4.7.4.6 T114N variant Ramachandran Plot

4.8 Stability and flexibility analysis of wildtype NUSAP1, T114N variant and R434M variant:

Deformability Plot: It shows how flexible each amino acid position of a protein is during molecular motion analysis. The X-axis represents the atom index (amino acid position), and the Y-axis represents the deformability value. The higher the peak, the easier it is for the amino acid residue to move; the lower the peak, the more rigid the amino acid residue.

The deformability value at position 434 decreases slightly from 0.528082 of wildtype to 0.491199 in the R434M variant. It suggests that the substitution of arginine with methionine slightly increases the rigidity and stability of the region. The deformability value at position 114 increases slightly from 0.348414 of wildtype to 0.399508 in the T114N variant. It suggests that the substitution of threonine with asparagine slightly increases the flexibility and stability of the region.

B-factor plot: A mobility plot that represents the flexibility and movement of amino acids within a protein structure. The X-axis represents the atom index (amino acid position), and the Y-axis represents the B-factor value. A higher B-factor value

indicates regions with greater flexibility and mobility, lower B-factor value indicates the structure is rigid.

For WT, R434M, and T114N, the overall protein structure is mostly conserved, showing a similar overall fold. In some regions, small local deviations indicate that, despite similar overall structures, the mutant proteins do have some differences in flexibility. Of the two mutants, R434M is less affected than T114N in terms of changes in B-factor values, suggesting a greater effect in terms of protein movement and flexibility.

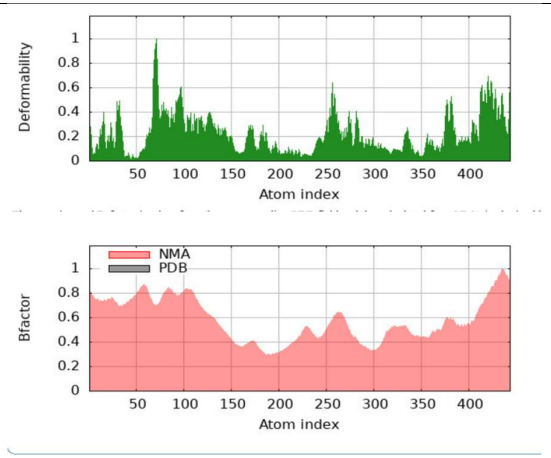
Eigenvalue plot: An eigenvalue plot is used to describe the stiffness of molecular motions. The lower the eigenvalue, indicates a flexible the region, whereas a high eigenvalue indicates a rigid structure. The X-axis represents normal mode numbers, where each mode indicates a specific type of protein movement, and the Y-axis represents the eigenvalue magnitude, which indicates the energy required for that movement.

The wild-type protein has the lowest first eigenvalue (7.136×10^{-10}), suggesting that it is more flexible in its conformational variation than the mutant proteins, with the lowest energy barrier for its dominant global motion. For the R434M variant, the first eigenvalue (7.850×10^{-10}) corresponds to a ~10% stiffer global motion compared to the wildtype protein. The T114N variant has the smallest first eigenvalue (1.119×10^{-9}), corresponding to a ~57% increase in protein stiffness, with elevated eigenvalues across all twenty modes studied.

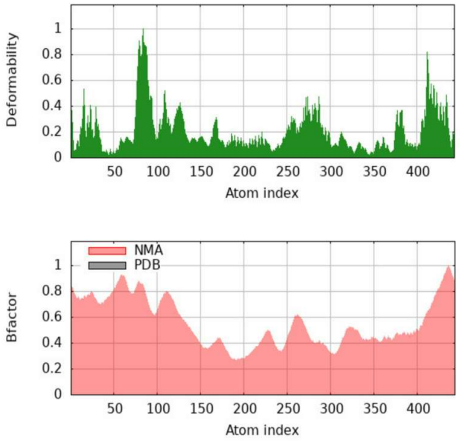
Variance plot: The variance plot displays the amount of protein motion present in the first twenty normal modes of protein. The normal modes are arranged from collective to localized motion and are plotted in order of mode number (1–20) on the X- axis and the percentage of total variance present in each mode on the Y-axis. The variance for each mode is represented by a purple bar, while the cumulative variance up to that mode is shown by a green bar.

For the wildtype protein, mode 1 contributes nearly 65% of the total motion, showing that most of the functional movement is dominated by a single major mode. The R434M variant, the contribution of mode 1 decreases by ~55%, with the protein motion being distributed over modes 2-8. The T114N variant shows a more evenly spread variance distribution plot with mode 1 contributing ~50% of the total motion.

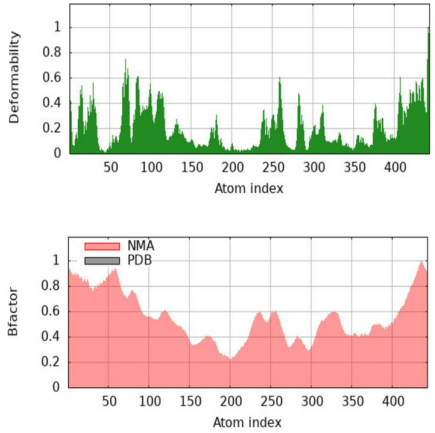
Table 4.8.1 iMODs Results of Wildtype, R434M mutant, and T114N mutant



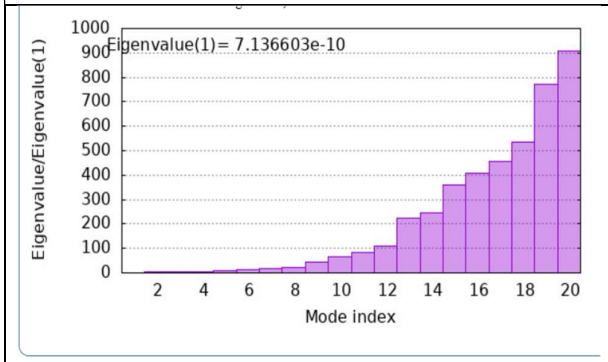
Deformability plot & B-factor plot of WT



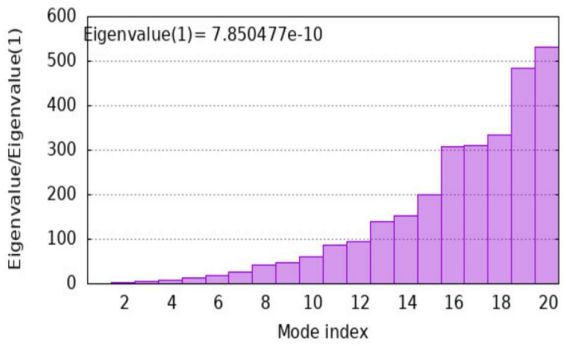
Deformability plot & B-factor plot of R434M



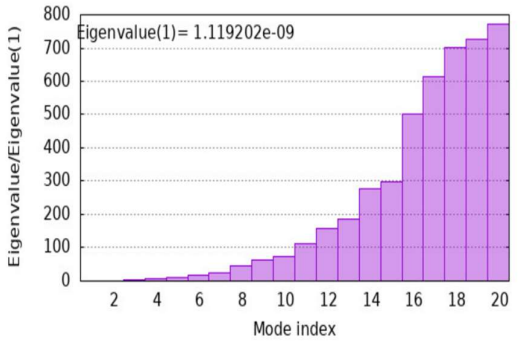
Deformability plot & B-factor plot of T114N



Eigenvalue plot of WT



Eigenvalue plot of R434M



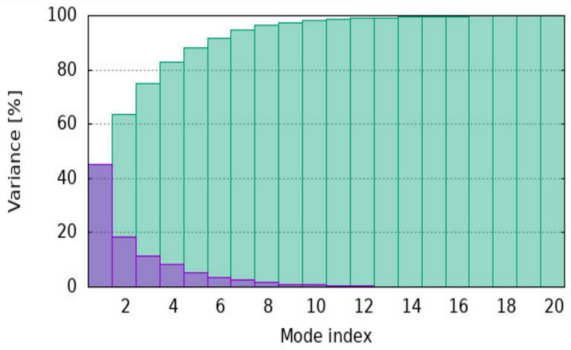
Eigenvalue plot of T114N

Covariance plot: It shows how different residues of a protein move relative to each other during molecular motion analysis. The x-axis and y-axis represent the residue position of the proteins. The intensity of color indicates the strength of correlation between residues, red region indicates two residues move in the same direction, blue region indicates two residues move in opposite directions, and white region indicates no relationship between residues.

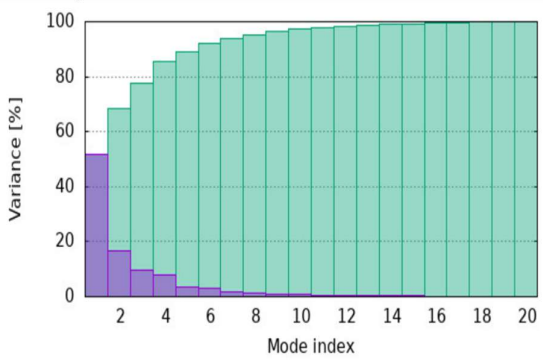
For the wildtype protein, the covariance map shows the most organized and different residues of the protein move together in a correlated motion. The R434M variant shows a decreased amount of correlated motion within the residues ~0–150 and ~300–450. Large blue anti-correlated patches in the R434M variant replaced the large off-diagonal red patches in the wildtype. In T114N, more changes in the covariance plot were caused by the loss of a hydrogen bond of residue 114. Strongly connected movements seen in WT and R434M are broken in the T114N variant.

Elastic network model: The Elastic Network Model (ENM) contact maps show the distances and interactions between all Cα atoms on the protein. The x- and y-axis represent the atom index going from the N-terminus (residue 0) to the C-terminus (residue ~430), and the color scale represents from black (very close and strong interactions) to grey for intermediate distances and white for indicating residues that are not interacting. The thick black main diagonal line represents interactions between neighboring residues in the protein chain, and dark spots away from the diagonal line are residues that are far apart in sequence but located close together and interact with each other, suggesting that the mutation does not significantly affect the overall structure.

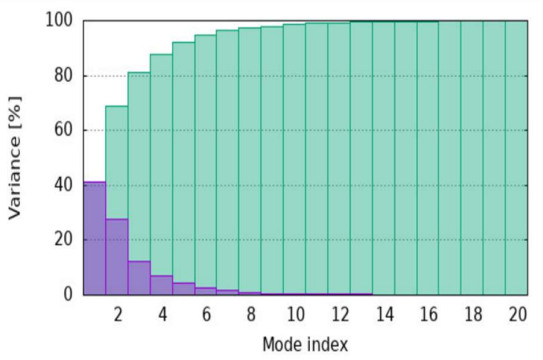
The ENM contact map for the R434M variant is very similar to that of the wild-type protein, which doesn't significantly affect the overall structure. The T114N variant maintains most of the overall structure of the wild-type protein, but it shows few contact regions in the N-terminal, which indicates that the mutation at residue 114 causes local small structural changes in the protein.



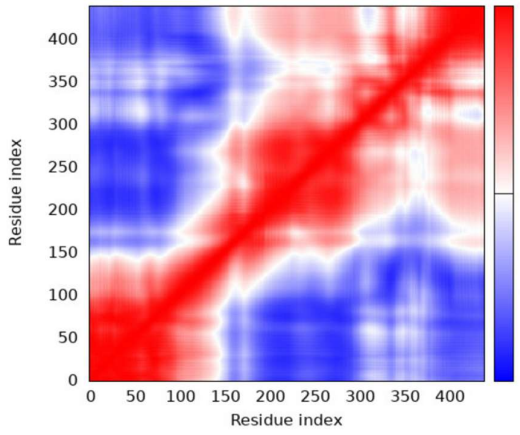
Variance plot of WT



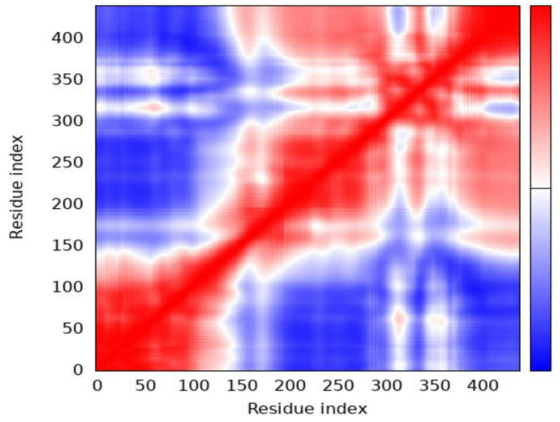
Variance plot of R434M



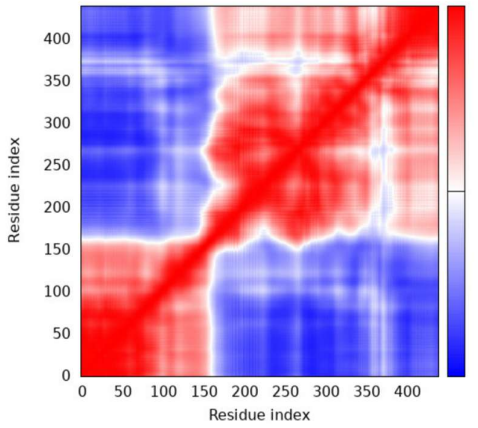
Variance plot of T114N



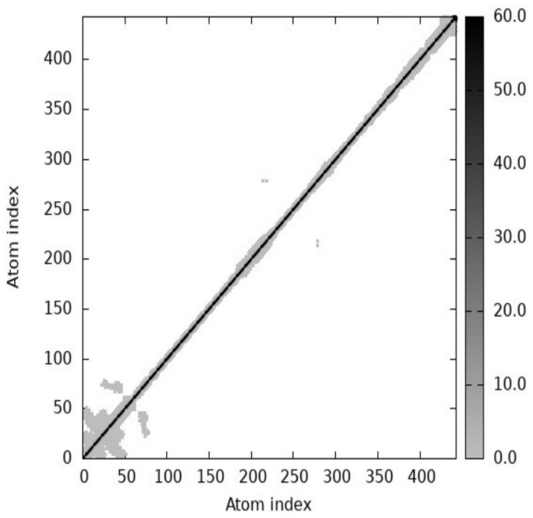
Covariance plot of Wildtype



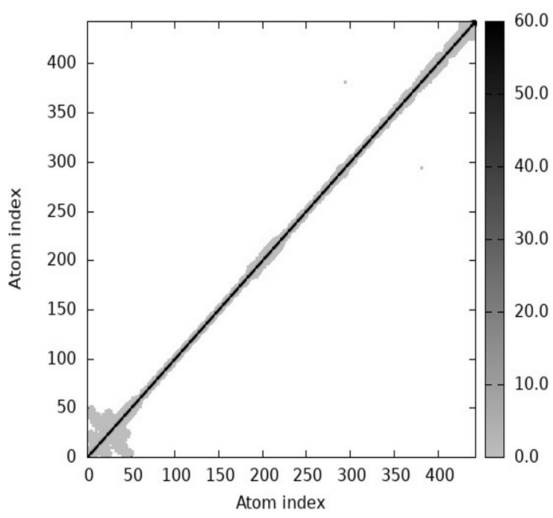
Covariance plot of R434M



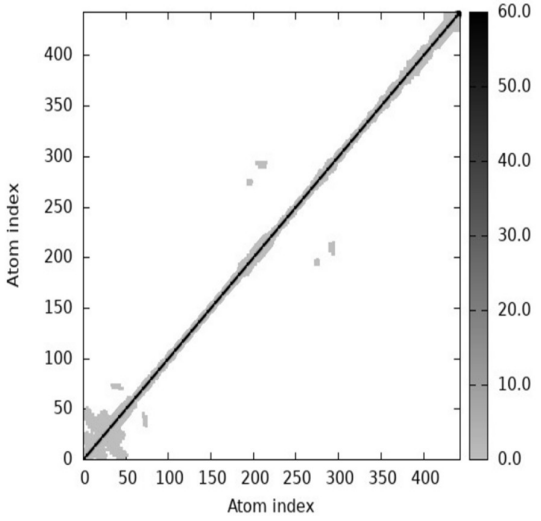
Covariance plot of T114N



Elastic network model of WT



Elastic network model of R434M

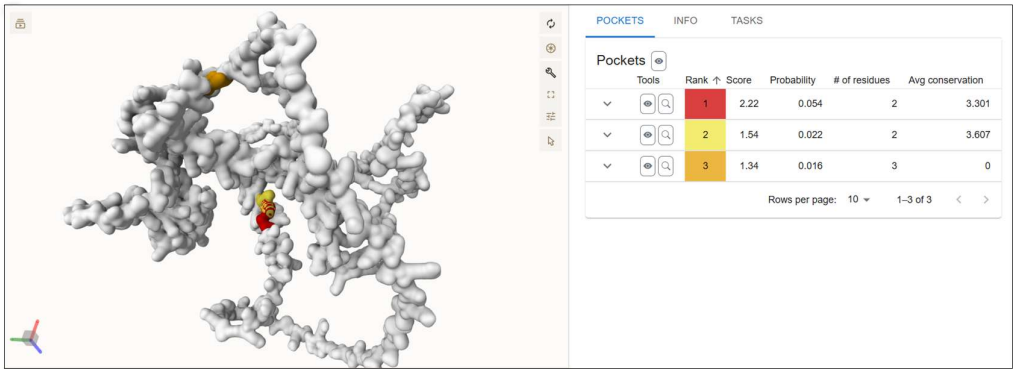


Elastic network model of T114N

Comparison of the WT, R434M, and T114N proteins by normal mode analysis (NMA) and covariant analysis) revealed that the most stable and with the most conserved native dynamic profile is the WT protein with a 'perfect' balance of flexibility and stability. It has strongly correlated residue movements, efficient communication between residues, and requires the least amount of energy for conformational changes (eigenvalue: 7.136×10^{-10}). Its residue interactions are very stable and form a solid, native protein structure that is preserved by both NMA and ENM analysis. The effects of mutation on the dynamics of the protein structure were moderate for R434M. The native dynamic profile of the WT protein was only slightly changed, with increased stiffness and lower flexibility of the R434M mutant protein. In contrast, the T114N mutant exhibited a number of changes in protein dynamics, including increased local flexibility, increased global stiffness, low correlated motion between residues, and structural changes, particularly around the site of mutation. The ranking of the preservation of native dynamics based on the normal mode analysis in combination is: WT > R434M > T114N. The WT protein keeps the most structure and dynamics conserved, which are needed for "normal" protein function.

4.9 Binding pockets identified for NUSAP1:

Binding pockets are predicted using PrankWeb by providing the input structure of the refined structure of wild-type NUSAP1, structures that were generated using AlphaFold.



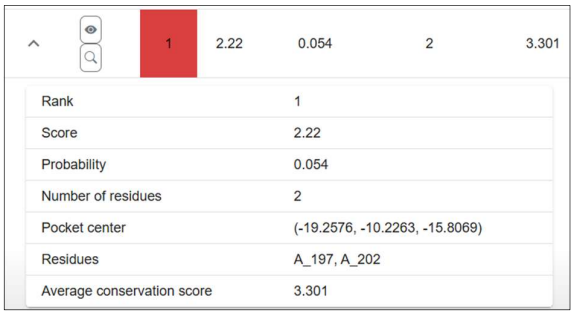


Fig. 4.9.1 Predicted binding Pockets of the NUSAP1 wild-type protein

Legend: 3D representation of protein structure is shown in white, in color, and binding pockets in red, yellow, and orange, indicating red means the deepest and most favorable pocket, yellow color indicates moderate binding suitability, and orange color indicates where ligand interaction is weaker compared to the red region.

There are three possible places for a ligand to bind to this protein structure. Based on the binding pocket analysis, the highest score for a predicted binding site is for Pocket 1. This is the highest score for a predicted binding site (2.22), and also the highest probability for a binding site (0.054). This pocket contains two residues: A197 and A202. The residue conservation for this pocket is moderate (3.301). While the residues in Pocket 2 have slightly higher conservation than those in Pocket 1, its binding score and probability are low, and thus it is not as good a candidate for binding as Pocket 1. The remaining potential sites were not considered for molecular docking and virtual screening studies because of their low binding scores and predictions for binding sites.

4.10.1 Virtual Screening of NUSAP1 gene using Mcule library:

The NUSAP1 protein was prepared by removing water molecules, adding hydrogen atoms, and adding Gasteiger charges. By providing the NUSAP1 prepared structure as a receptor for input to the Mcule chemical compound library, predicted binding sites coordinates were identified using PrankWeb. The coordinates were set to x = y = z = 20 A^{o} with a center at x = -19.2576, y = -10.2263, and z = -15.8069. It revealed the 100 compounds with docking scores, SMILES notation, and molecular descriptors of chemical compounds; the results were downloaded for further analysis.

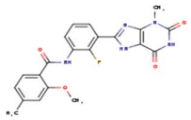
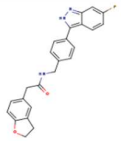
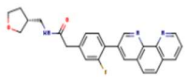
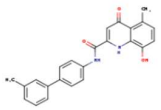
The downloaded results were filtered based on Lipinski's rule of five, which is used for the evaluation of the oral drug likeness of small molecules. Compounds with molecular weight ≤ 500 g/mol, LogP ≤ 5, hydrogen bond donors ≤ 5, and hydrogen

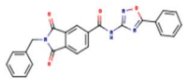
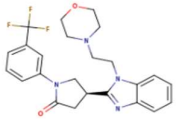
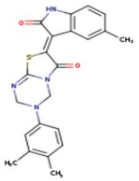
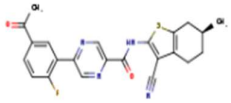
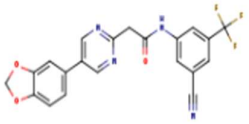
bond acceptors ≤ 10 are considered good oral drug candidates. Bioavailability score predicts how much of a drug can enter the bloodstream after oral intake, where a value around 0.55 is considered acceptable. Lipophilicity is the ability of a chemical compound to dissolve in lipids or nonpolar solvents, and is expressed using the parameter (LogP), which measures how much a substance dissolves in fats or nonpolar solvents as compared to water. LogP < 0 indicates high hydrophilic and low lipophilicity, logP (1-3) balanced lipophilicity and hydrophilicity, and it is an ideal range for oral drugs. LogP (3 – 5) indicates poor solubility, and log > 5 indicates highly lipophilic. Gastrointestinal (GI) absorption predicts how well a compound can be absorbed in the digestive tract after oral intake. High GI suggests that the compound is well absorbed after oral intake. Low GI suggests poor absorption in the digestive tract. Blood–brain barrier (BBB) permeation predicts whether the compound can cross the blood-brain barrier and enter the brain. Pan-Assay Interference Compounds (PAINS) alerts identify compounds that give false positive results, while Brenk alerts identify the structural fragments of compounds that may cause toxicity. Compounds with zero PAINS and Brenk alerts are considered for drug development.

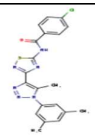
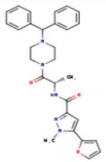
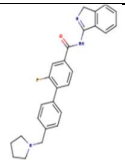
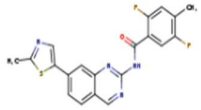
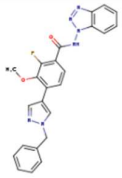
34 compounds were selected after filtering the chemical compounds based on Lipinski's rule, and the physicochemical properties were evaluated to consider a lead compound. 9 Mcule compounds were selected after filtering based on SWISSADME, and performed Molecular docking analysis using AutoDock Vina against the NUSAP1 protein and considering the Mcule chemical compounds as ligands. Binding site coordinates were identified using PrankWeb; the coordinates were x = y = z = 20 A^o with a center at x = -19.2576, y = -10.2263, and z = -15.8069.

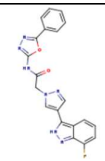
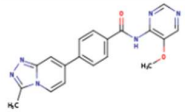
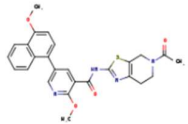
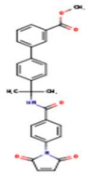
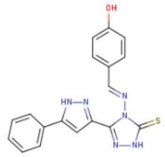
2D interaction analysis shows the types of interactions between the ligand and amino acid residues, which helps identify residues involved in binding and verifies toxicity using the Mcule built-in toxicity checker tool. 4 non- toxic lead compounds to predict the toxicity using ProTox 3.0. Toxicological parameters include oral toxicity, organ toxicity (hepatotoxicity, nephrotoxicity, neurotoxicity, cardiotoxicity, immunotoxicity), carcinogenicity, cytotoxicity, and nutritional toxicity.

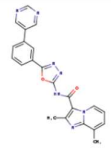
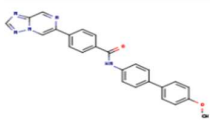
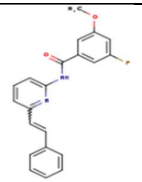
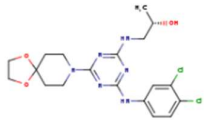
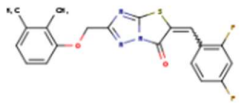
Table 4.10.1.1 Mcule & SWISSADME results of NUSAP1 Wildtype

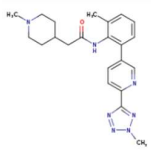
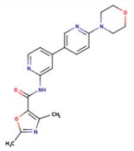
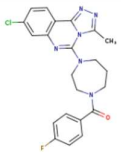
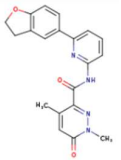
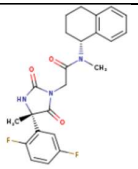
Sr.no	Mcule ID & Structure	Docking scores	SwissADME results						
			Log P	Bio-availability score	GI absorption	BBB-permeant	PAINS	Brenk Alerts	Lipinski violation
1	 MCULE-1728327632-0-1	-6	2.57	0.55	High	No	0	0	0
2	 MCULE-3003239086-0-1	-5.9	4.12	0.55	High	Yes	0	0	0
3	 MCULE-2542073912-0-1	-5.8	3.96	0.55	High	No	0	1	0
4	 MCULE-2272854841-0-1	-5.8	4.02	0.55	High	No	0	0	0

5	 MCULE-1680887574-0-1	-5.7	3.19	0.55	High	No	0	1 phthalimide	0
6	 MCULE-5779512393-0-1	-5.6	3.47	0.55	High	Yes	0	0	0
7	 MCULE-2105125611-0-2	-5.6	3.08	0.55	High	No	1 anil_di_alk_E	0	0
8	 MCULE-3722816089-0-1	-5.6	4.05	0.55	High	No	0	0	0
9	 MCULE-1197434216-0-1	-5.6	3.41	0.55	High	No	0	0	0

10	 MCULE-4965991575-0-1	-5.5	4.04	0.55	High	No	0	0	0
11	 MCULE-7220877219-0-1	-5.5	3.10	0.55	High	No	0	0	0
12	 MCULE-2196815405-0-1	-5.5	4.54	0.55	High	No	0	0	1 LogP>4.15
13	 MCULE-6557310118-0-1	-5.5	4.50	0.55	High	No	0	0	0
14	 MCULE-5422523449-0-1	-5.5	3.52	0.55	High	No	0	0	0

15	 MCULE-3428911210-0-1	-5.5	2.50	0.55	High	No	0	0	0
16	 MCULE-4544406282-0-1	-5.5	2.24	0.55	High	No	0	0	0
17	 MCULE-7224783834-0-1	-5.5	3.55	0.55	High	No	0	0	0
18	 MCULE-7063992425-0-1	-5.4	3.82	0.55	High	No	0	1 phthalimide	0
19	 MCULE-9571518171-0-2	-5.4	2.99	0.55	High	No	1 hzone_phenol_B	2 imine_1, thiocarbonyl_group	0

20	 MCULE-3298531077-0-1	-5.4	2.89	0.55	High	No	0	0	0
21	 MCULE-7854802180-0-1	-5.4	3.70	0.55	High	No	0	0	0
22	 MCULE-4271300701-0-1	-5.4	3.99	0.55	High	Yes	0	0	0
23	 MCULE-5250547885-0-1	-5.3	2.75	0.55	High	No	0	0	0
24	 MCULE-9657453300-0-1	-5.3	4.48	0.55	High	No	0	0	0

25	 MCULE-2586617930-0-1	-5.3	2.65	0.55	High	No	0	0	0
26	 MCULE-7477634163-0-1	-5.3	2.00	0.55	High	No	0	0	0
27	 MCULE-6724108984-0-1	-5.3	3.55	0.55	High	Yes	0	0	0
28	 MCULE-9488734308-0-1	-5.3	2.33	0.55	High	No	0	0	0
29	 MCULE-9778710119-0-2	-5.3	3.05	0.55	High	Yes	0	1 hydantoin	0

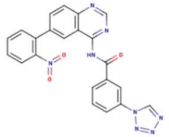
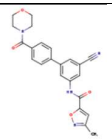
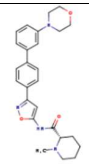
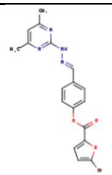
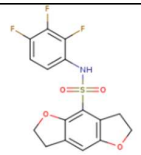
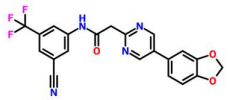
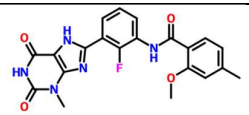
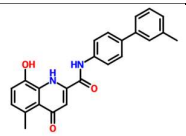
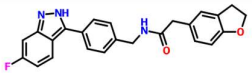
30	 MCULE-2828455476-0-1	-5.3	2.54	0.55	High	No	0	1 nitro group	1 Oxygen donors >10
31	 MCULE-5047181802-0-1	-5.3	2.49	0.55	High	No	0	0	0
32	 MCULE-1123785341-0-1	-5.3	3.30	0.55	High	Yes	0	0	0
33	 MCULE-7714268730-0-1	-5.2	3.68	0.55	High	No	0	2 imine_1, phenol_ester	0
34	 MCULE-8915939307-0-1	-5.2	3.33	0.55	High	No	0	1 halogenated_ ring_2	0

Table 4.10.1.2 2D-Interactions of chemical compounds with Wildtype NUSAP1

Structure & MCULE ID	Binding energy	2D Protein-Ligand Interactions					
		Van der Waals	Carbon-hydrogen bond	π-π stacking	π-Alkyl	Halogen (Fluorine)	Conventional hydrogen bonds
 MCULE-1197434216-0-1	-6.029 kcal/mol	LYS207, MET196	ILE199	TYR202	LYS206, ILE203	SER198, GLU197	-
 MCULE-1728327632-0-1	-6.626 kcal/mol	LYS207, SER198, ASP200	GLN201	-	ARG205	-	GLU211, GLU204
 MCULE-2272854841-0-1	-6.802 kcal/mol	MET196, SER198	GLN201	TYR202	ILE203, ILE199	-	-
 MCULE-3003239086-0-1	-7.628 kcal/mol	GLU197	-	TYR202, PHE210	LYS206, ILE199, LYS207	-	ILE203

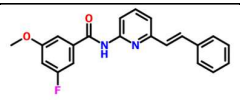
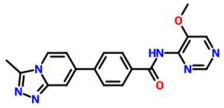
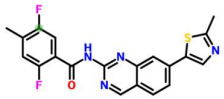
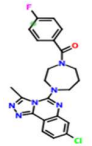
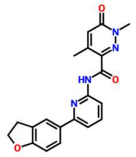
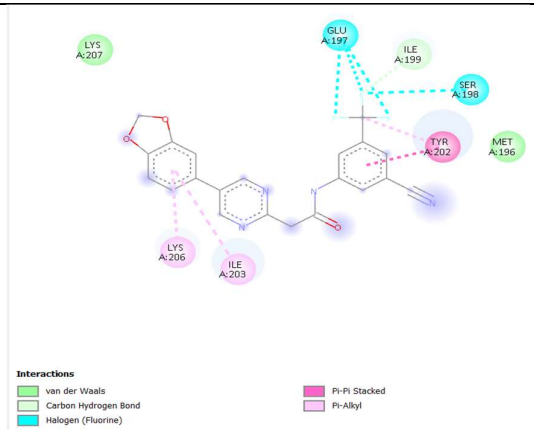
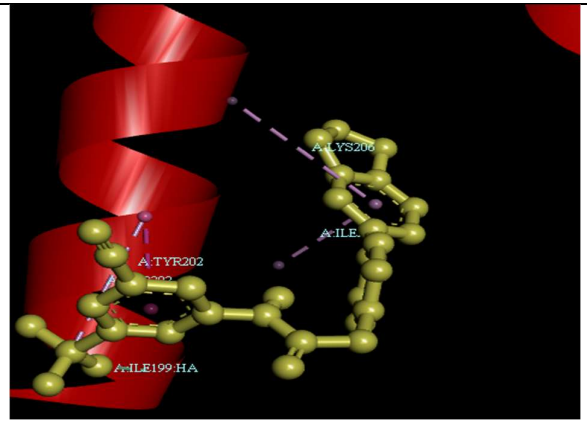
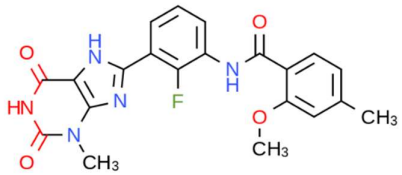
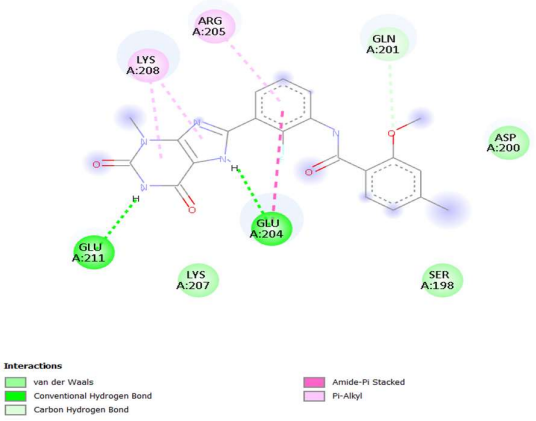
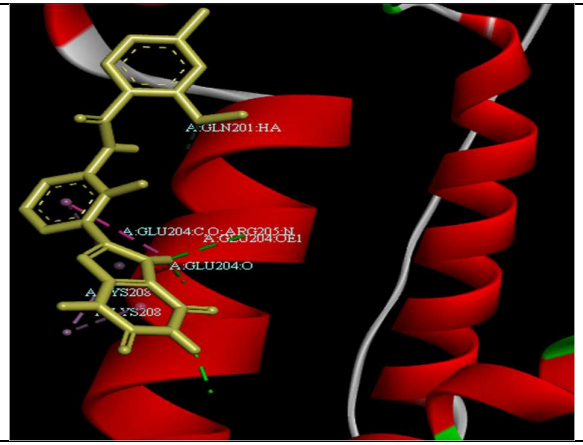
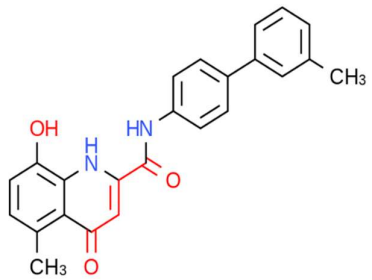
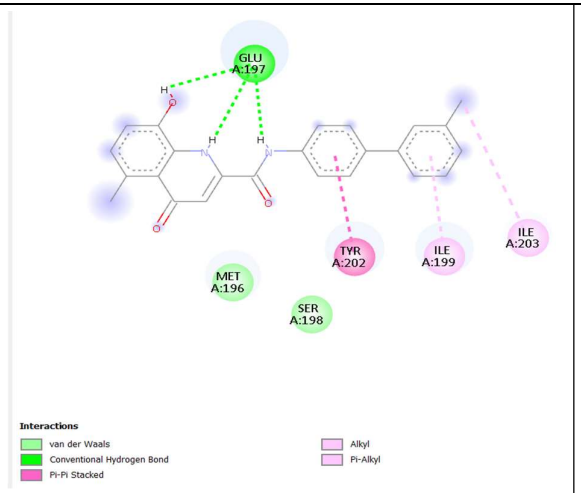
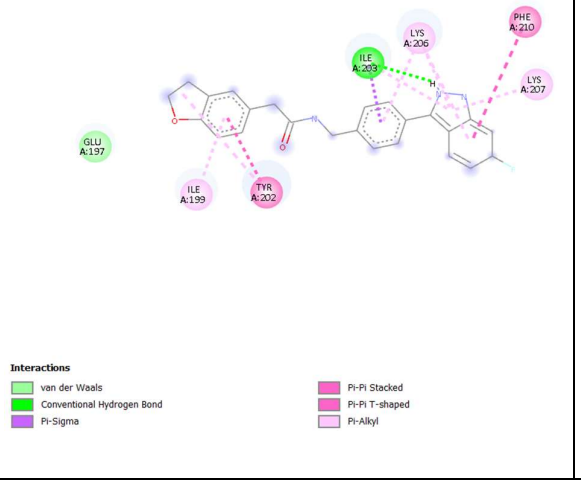
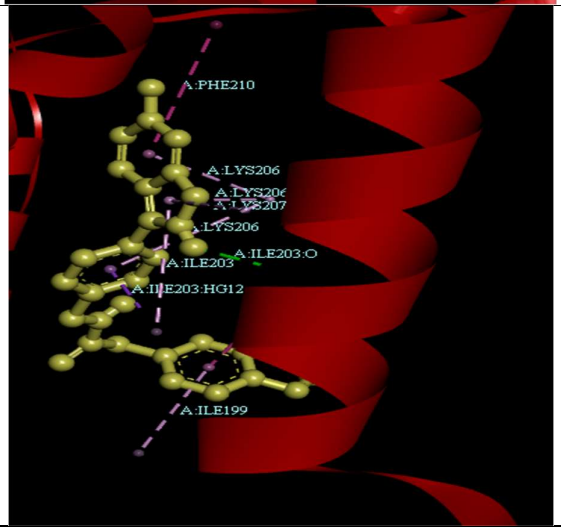
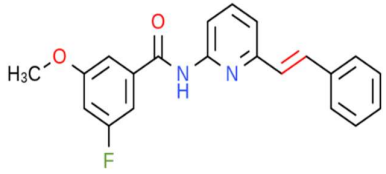
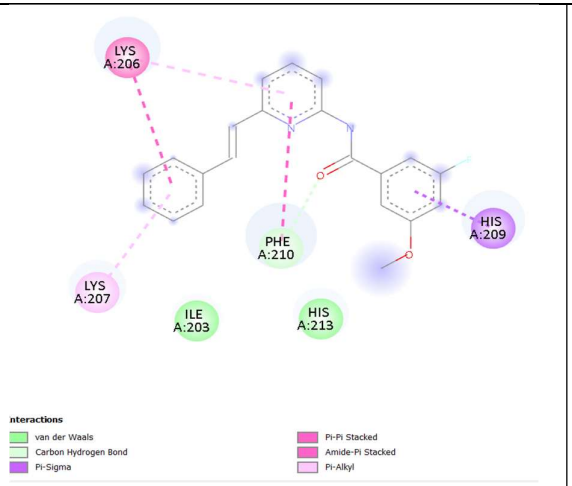
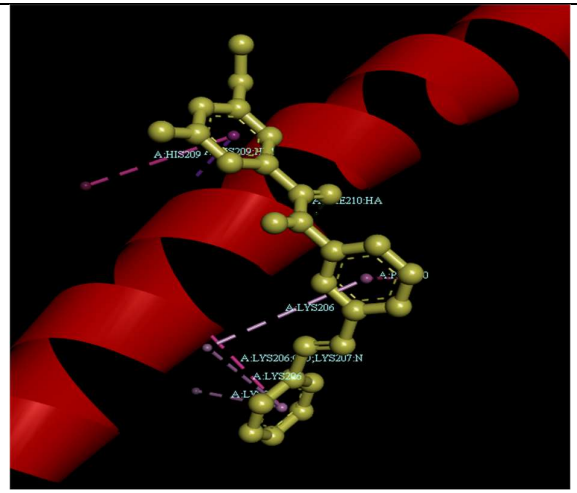
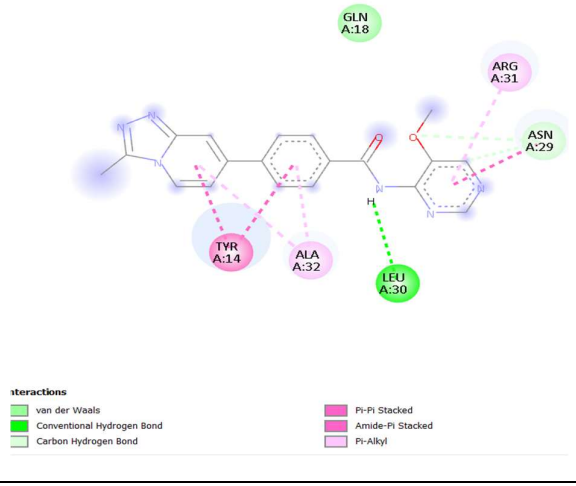
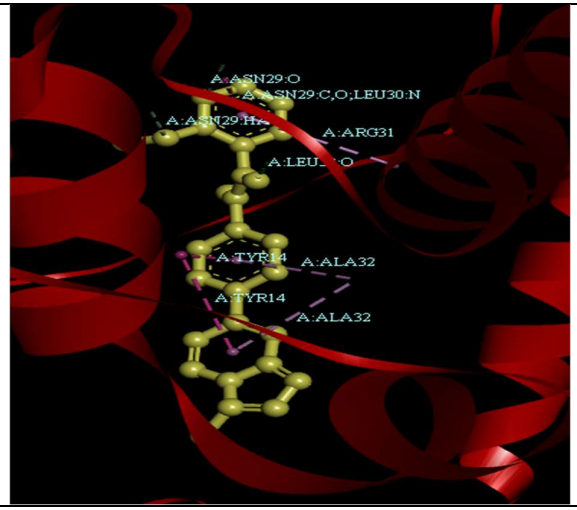
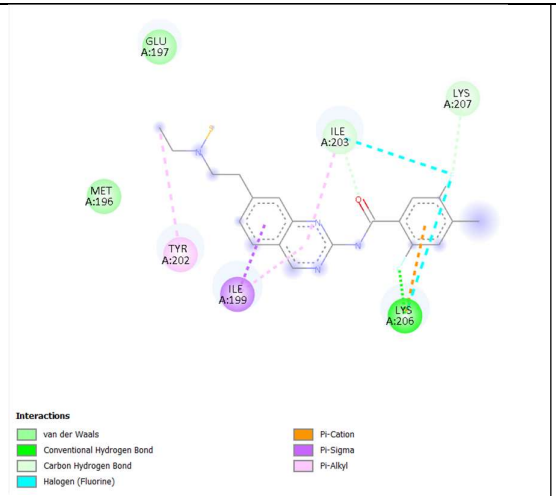
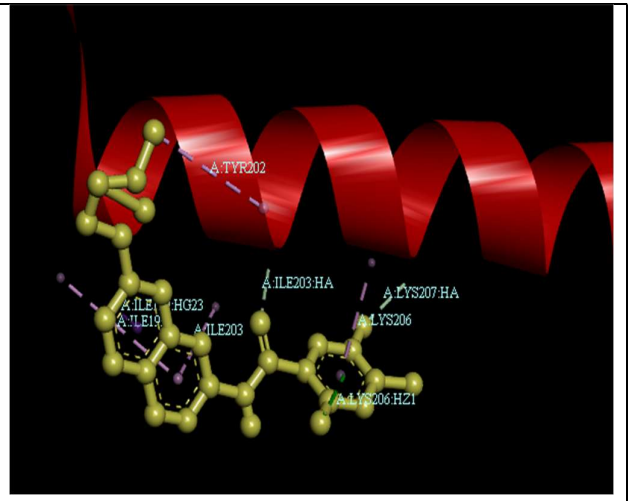
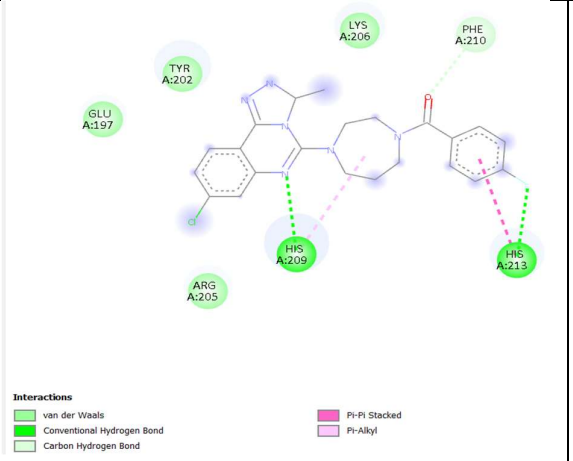
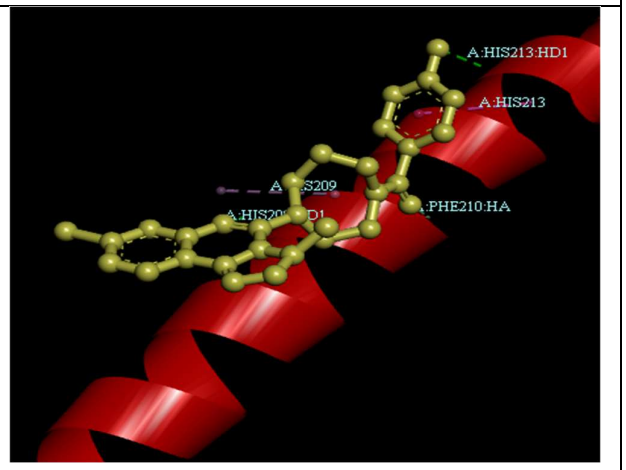
 MCULE-4271300701-0-1	-6.509 kcal/mol	HIS213, ILE203	PHE210	LYS206	LYS207	-	-
 MCULE-4544406282-0-1	-6.483 kcal/mol	ASN29, GLN18	-	TYR14	ARG31, ALA32	-	LEU30
 MCULE-6557310118-0-1	-6.826 kcal/mol	MET196, GLU197	LYS206	-	TYR202	-	-
 MCULE-6724108984-0-1	-6.479 kcal/mol	TYR202, GLU197, LYS06	HIS209, HIS213	-	-	-	PHE210, ARG205
 MCULE-9488734308-0-1	-7.171 kcal/mol	GLU197, ASP200	MET196, PHE193	-	LYS194	-	-

Table 4.10.1.3 Wildtype NUSAP1-Ligand Interactions

Sr	TOXICITY CHECKER	2D-INTERACTIONS	3D-INTERACTIONS
1	Toxicity Checker: FAIL  MCULE-1197434216-0-1		
2	Toxicity Checker: FAIL  MCULE-1728327632-0-1		

3	Toxicity Checker: FAIL  MCULE-2272854841-0-1	 	
4	Toxicity Checker: OK MCULE-3003239086-0-1	 	

5	Toxicity Checker: FAIL  MCULE-4271300701-0-1		
6	Toxicity Checker: OK MCULE-4544406282-0-1		

7	Toxicity Checker: **OK** MCULE-6557310118-0-1		
8	Toxicity Checker: OK MCULE-6724108984-0-1		

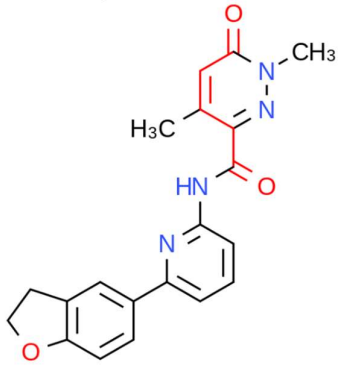
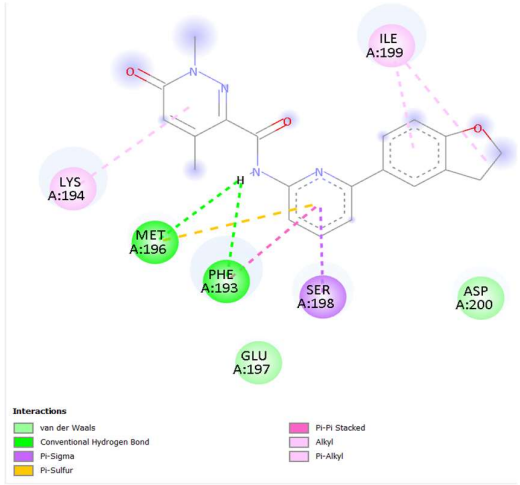
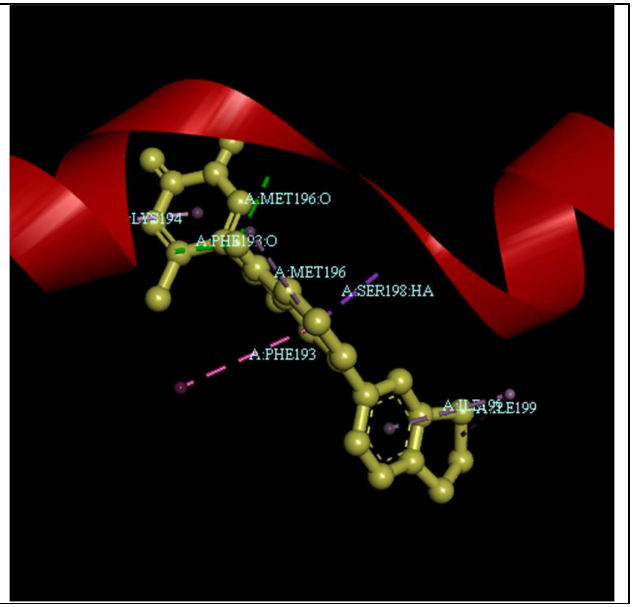
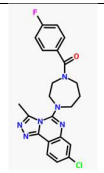
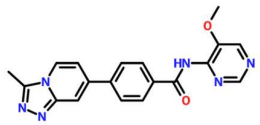
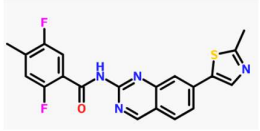
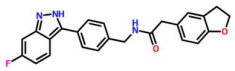
9	Toxicity Checker: FAIL  MCULE-9488734308-0-1	 	

Table 4.10.1.4 ProTox results of chemical compounds

Sr	Compounds	Oral Toxicity	Hepatotoxicity	Neurotoxicity	Nephrotoxicity	Lung toxicity	Cardiotoxicity	Carcinogenicity	Immunotoxicity	Cell toxicity	Clinical toxicity	Nutritional toxicity
1	 PubChem CID: 46063666	Low toxicity	Not harmful	Harmful	Not harmful	Harmful	Not harmful	Not harmful	Harmful	Not harmful	Possible toxicity	Not harmful
		- No possible binding to toxicity targets. - SMILES: C1(C(=O)N2CCN(CCC2)C2N3C(C)=NN=C3C3=C(N=2)C=C(Cl)C=C3)C=CC(F)=CC=1 - Compound Name: 1-{8-Chloro-3-methyl-[1,2,4]triazolo[4,3-C]quinazolin-5-YL}-4-(4-fluorobenzoyl)-1,4-diazepane										
2		Low toxicity	Harmful	Harmful	Not harmful	Harmful	Not harmful	Not Harmful	Not Harmful	Not harmful	Not harmful	Not Harmful
		- induced Thyroid hormone receptor beta (THRβ) event - No possible binding to toxicity targets. - SMILES: C1(C=CC(C(=O)NC2N=CN=CC=2OC)=CC=1)C1C=CN2C(=NN=C2C)C=1										
3		Low toxicity	Harmful	Harmful	Not harmful	Not harmful	Not harmful	Not harmful	Not harmful	Not harmful	Harmful	Not harmful
		- No possible binding to toxicity targets. - SMILES: C1(SC(C)=NC=1)C1=CC=C2C(N=C(N=C2)NC(C2C(F)=CC(C)=C(F)C=2)=O)=C1										
4		Low toxicity	Harmful	Harmful	Not harmful	Harmful	Not harmful	Harmful	Harmful	Not harmful	Harmful	Not Harmful
		- No possible binding to toxicity targets. - SMILES: C1(C=CC(CNC(CC2C=C3CCOC3=CC=2)=O)=CC=1)C1=C2C(C=C(C=C2)F)=NN1										

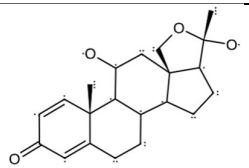
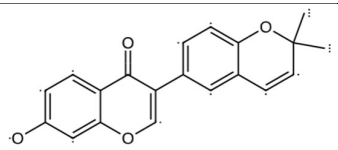
Docking score (score generated automatically in mcule) and binding energy (score generated while performing molecular docking) represent a value generated during molecular docking that estimates how strongly NUSAP1 binds with mcule chemical compounds. The MCULE-3003239086-0-1 complex forms one conventional hydrogen bond with the Ile203 atom, with a binding energy of -7.628 Kcal/mol and docking score of -5.9 Kcal/mol. The MCULE-4544406282-0-1 complex forms one conventional hydrogen bond with the Leu30 atom as well as a carbon-hydrogen bond with the Asn29 atom, with a binding energy of -6.483 Kcal/mol and docking score of -5.5 Kcal/mol. The MCULE-6557310118-0-1 complex forms the highest number of hydrogen bonds to the atoms of the binding site. One conventional hydrogen bond is formed with the Lys206 atom, while the complex also forms two carbon-hydrogen bonds with Ile203 and Lys207 atoms, with a binding energy of -6.826 Kcal/mol and docking score of -5.5 Kcal/mol. The MCULE-6724108984-0-1 complex forms two conventional hydrogen bonds with the His209 and His213 atoms, as well as a carbon-hydrogen bond with the Phe210 atom, with a binding energy of -6.479 Kcal/mol and docking score of -5.3 Kcal/mol.

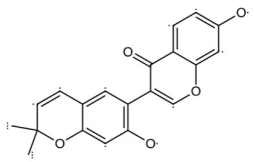
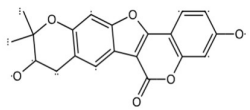
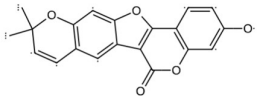
4.10.2 Virtual Screening of NUSAP1 gene using IMPPAT Database:

A total of ~17000 phytochemical compounds of Indian medicinal plants were retrieved from the IMPPAT 2.0 database. The compounds were retrieved in .sdf format. Using Obabel generated 3d conformers and performing energy minimization, applied Lipinski rule filtering for evaluating the oral drug likeness of phytocompounds.

10638 phytochemical compounds satisfied the Lipinski rule and performed molecular docking studies with NUSAP1 protein using AutoDock Vina for predicting protein-ligand binding affinities. Before performing docking, the NUSAP1 protein was prepared by removing water molecules, adding hydrogen atoms, and adding Gasteiger charges. Predicted binding sites identified in grid box coordinates using PrankWeb. The grid box dimensions were set to x = y = z = 20 A^{o} with a center at x = -19.2576, y = -10.2263, and z = -15.8069. Based on docking results, the top 100 phytochemicals with the lowest binding energies were selected as lead compounds for further analysis. The top 100 lead compounds identified from molecular docking were submitted to SwissADME for evaluation of their Absorption, Distribution, Metabolism, and Excretion properties and overall drug likeness.

Table 4.10.2.1 2D Interactions of phytocompounds with NUSAP1 from IMPPAT database

Sr.No	Compounds	Binding energy	2D-Interactions						
			Van der waals	Conventional hydrogen bond	Carbon-Hydrogen bond	π -Alkyl	π-Sigma	π- π- T-shaped	π- π-Stacked
1	 SMILES: O1[C][C@]23[C]([C]4[C]([C]([O])[C]3)[C@@]3(C(=[C]C(=O)[C]=[C]3)[C][C]4)[C])[C][C][C]2[C@]1([O])[C]	-7.084 Kcal/mol	MET196, SER198, ILE203	GLU197	ILE199	TYR202	-	-	-
2	 SMILES: O1C([C]=[C]C2=[C]C(=[C][C]=C12)C1=[C]OC2=[C]C(=[C][C]=C2C1=O)[O])([C])[C]	-6.771 Kcal/mol	GLU197, TYR202, PHE210	-	ARG205	HIS209	LYS206	-	-

3	 SMILES: O1C([C]=[C]C2=[C]C(=C([O])[C]=C12)C1=[C]OC2=[C]C(=[C][C]=C2C1=O)[O])([C])[C]	-6.666 Kcal/mol	GLU197, PHE210	ARG205, TYR202	-	LYS206	-	HIS209	-
4	 SMILES: CC1(C(CC2=CC3=C(C=C2O1)OC4=C3C(=O)OC5=C4C=CC(=C5)O)O)C	-6.656 Kcal/mol	GLU197, PHE210, ARG205, LYS206	HIS209	-	-	-	TYR202	-
5	 SMILES: CC1(C=CC2=CC3=C(C=C2O1)OC4=C3C(=O)OC5=C4C=CC(=C5)O)C	-6.639 Kcal/mol	ILE203	-	-	ILE199, MET196	-	-	TYR202

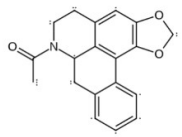
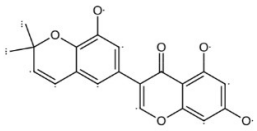
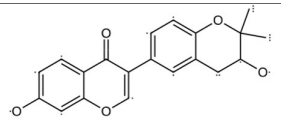
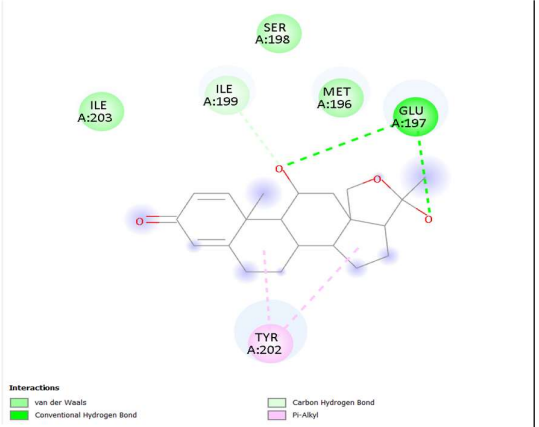
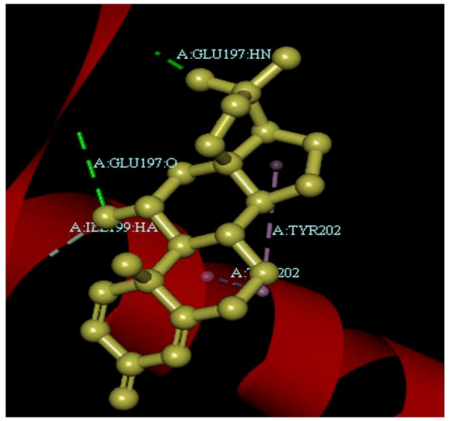
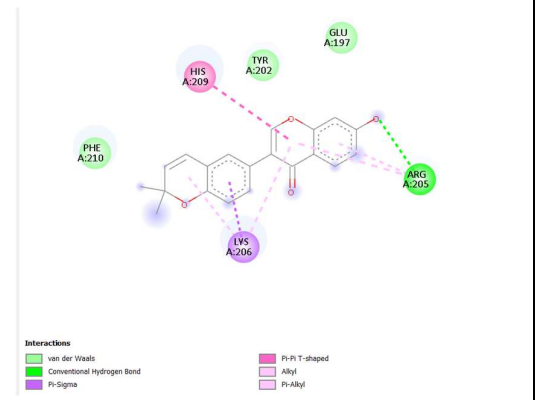
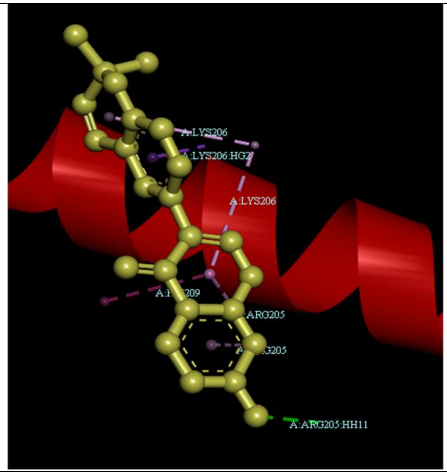
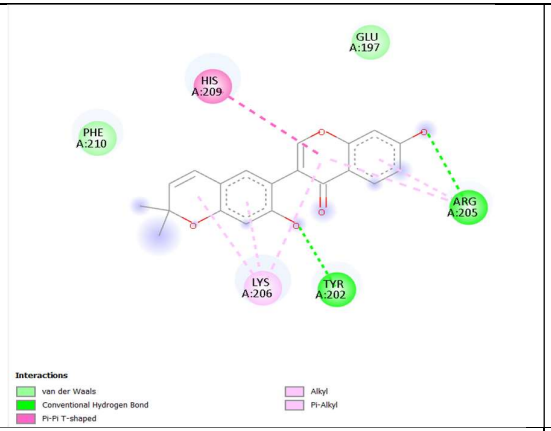
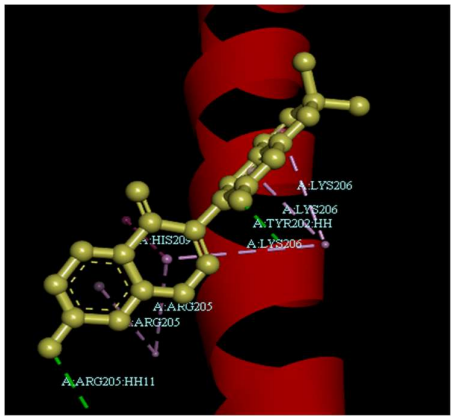
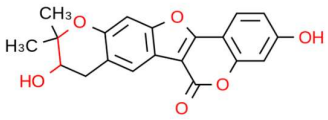
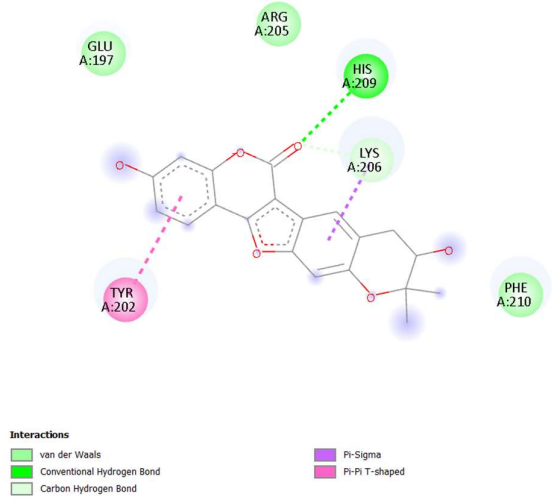
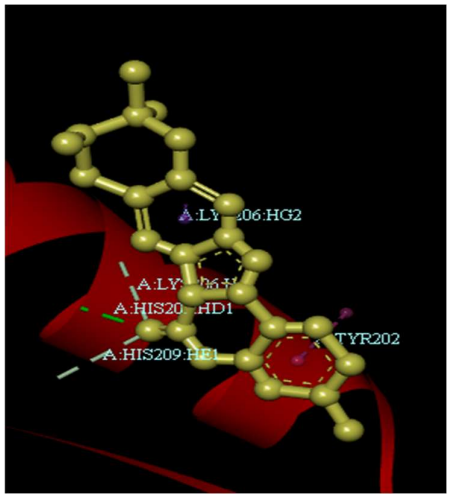
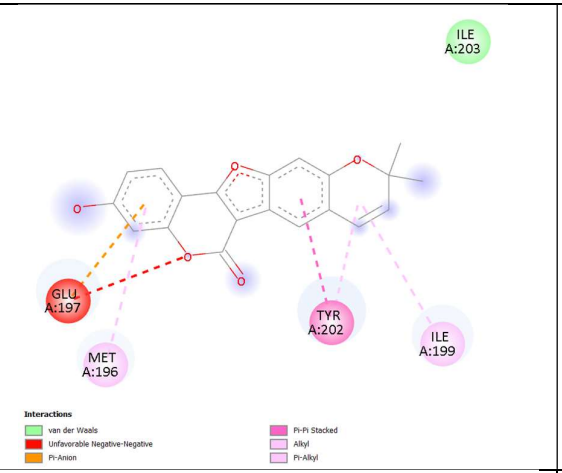
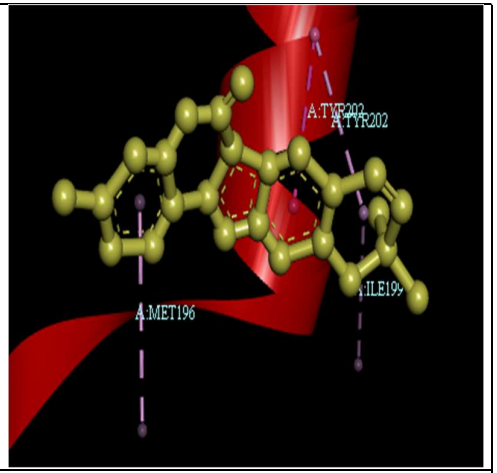
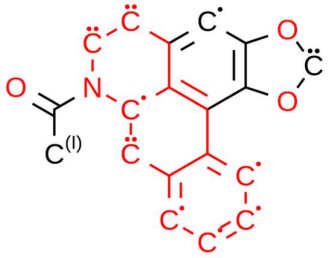
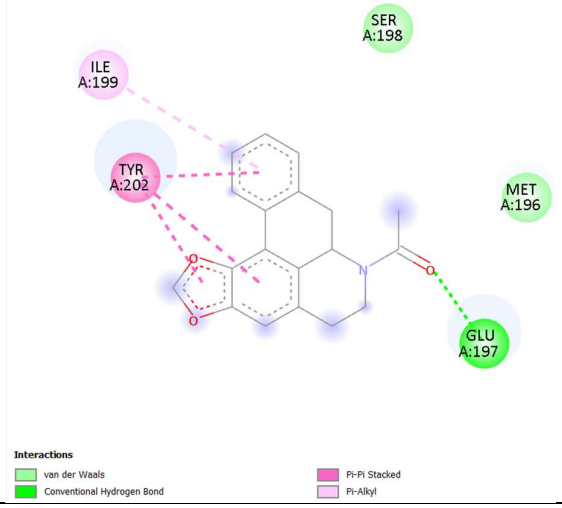
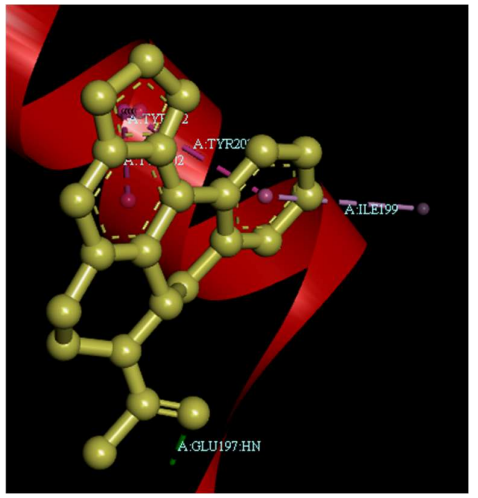
6	 SMILES: O1C2=C(O[C]1)[C]=C1C3=C2C2=[C][C]=[C][C]=C2[C][C]3N([C][C]1)C(=O)[C]	-6.559 Kcal/mol	SER198, MET196	GLU197	-	ILE199	-	-	TYR202
7	 SMILES: O1C([C]=[C]C2=[C]C(=[C]C(=C12)[O])C1=[C]OC2=[C]C(=[C]C(=C2C1=O)[O])[O])([C])[C]	-6.555 Kcal/mol	GLU197, TYR202, PHE210	ARG205	-	LYS206	-	HIS209	-
8	 SMILES: O1C([C]([O])[C]C2=[C]C(=[C][C]=C12)C1=[C]OC2=[C]C(=[C][C]=C2C1=O)[O])([C])[C]	-6.534 Kcal/mol	GLU197, TYR202, PHE210	ARG205	-	LYS206	-	HIS209	-

Table 4.10.2.2 Protein–Ligand Interaction Analysis of Phytocompounds with NUSAP1

SR	TOXICITY CHECKER	2D-INTERACTIONS	3D-INTERACTIONS
1	Toxicity Checker: OK SMILES: O1[C][C@]23[C]([C]4[C]([C]([O])[C]3)[C@@]3(C(=[C]C(=O)[C]=[C]3)[C][C]4)[C])[C][C][C]2[C@]1([O])[C] - 5004		
2	Toxicity Checker: OK SMILES: O1C([C]=[C]C2=[C]C(=[C][C]=C12)C1=[C]OC2=[C]C(=[C][C]=C2C1=O)[O])([C])[C] - 2911		

3	Toxicity Checker: OK SMILES: O1C([C]=[C]C2=[C]C(=C([O])[C]=C12)C1=[C]OC2=[C]C(=[C][C]=C2C1=O)[O])([C])[C]- 7987		
4	Toxicity Checker: FAIL  SMILES: CC1(C(CC2=CC3=C(C=C2O1)OC4=C3C(=O)OC5=C4C=CC(=C5)O)O)C - 4769		

5	Toxicity Checker: OK SMILES: CC1(C=CC2=CC3=C(C=C2O1)OC4=C3C(=O)OC5=C4C=CC(=C5)O)C - 1481		
6	Toxicity Checker: FAIL  SMILES: O1C2=C(O[C]1)[C]=C1C3=C2C2=[C][C]=[C][C]=C2[C][C]3N([C][C]1)C(=O)[C]-4443		

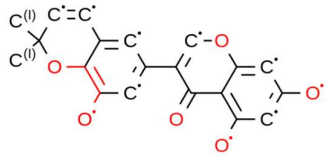
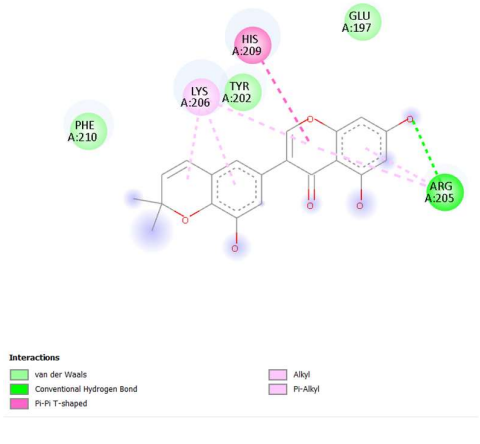
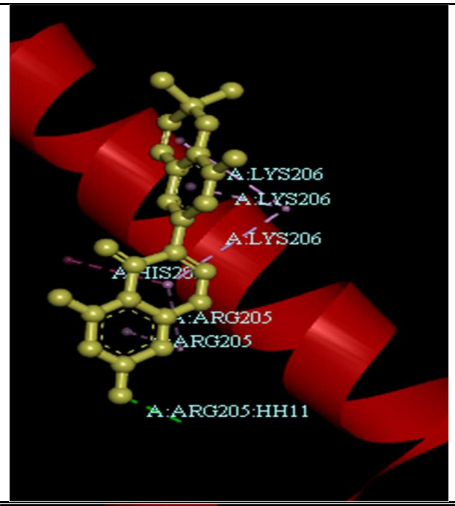
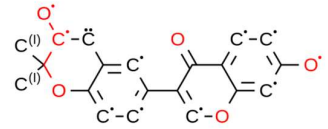
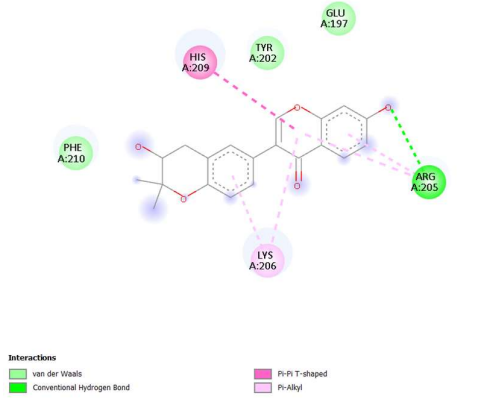
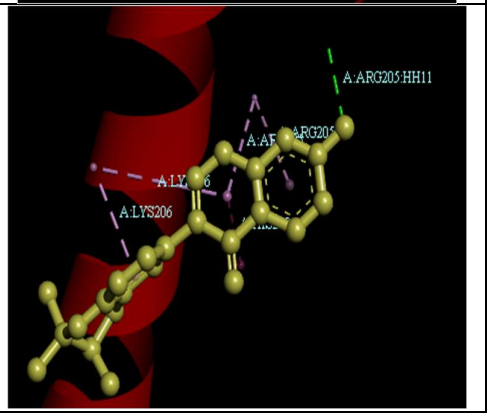
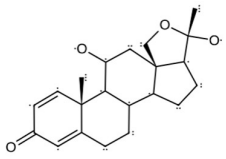
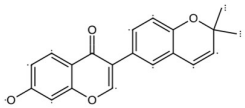
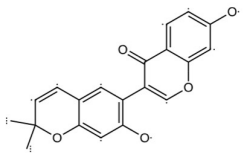
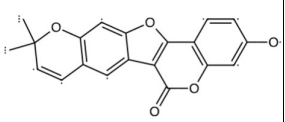
7	Toxicity Checker: FAIL  SMILES: O1C([C]=[C]C2=[C]C(=[C]C(=C12)[O])C1=[C]OC2=[C]C(=[C]C(=C2C1=O)[O])[O])([C])[C] - 5672		
8	Toxicity Checker: FAIL  SMILES: O1C([C]([O])[C]C2=[C]C(=[C][C]=C12)C1=[C]OC2=[C]C(=[C][C]=C2C1=O)[O])([C])[C] - 2556		

Table 4.10.2.3 ProTox results of Phytocompounds

Sr	Compounds	Oral Toxicity	Hepato toxicity	Neurot oxicity	Nephro toxicity	Lung toxicity	Cardio toxicity	Carcino genicity	Immuno toxicity	Cell toxicity	Clinical toxicity	Nutritio nal toxicity
1		Low toxicity	Not harmful	Not harmful	Not harmful	Harmful	Not harmful	Not harmful	Harmful	Harmful	Possible toxicity	Harmful
		- Blocks male hormones and binds to the androgen receptor site. - No possible binding to toxicity targets. - SMILES:O1[C][C@]23[C]([C]4[C]([C]([O])[C]3)[C@@]3(C(=[C]C(=O)[C]=[C]3)[C][C]4)[C])[C][C][C]2[C@]1([O])[C]										
2	 PubChem CID: 5316097	Low toxicity	Not harmful	Not harmful	Not harmful	Harmful	Not harmful	Harmful	Harmful	Not harmful	Not harmful	Harmful
		- Blocks female hormones and may affect cellular energy function. - PGH1(Prostaglandin G/H Synthase 1) has possible binding to toxicity targets. - SMILES: O1C([C]=[C]C2=[C]C(=[C][C]=C12)C1=[C]OC2=[C]C(=[C][C]=C2C1=O)[O])([C])[C] - Compound Name: Corylin										
3	 PubChem CID: 11244593	Low toxicity	Not harmful	Not harmful	Not harmful	Harmful	Not harmful	Not harmful	Harmful	Not harmful	Not harmful	Harmful
		- Blocks female hormones and may affect cellular energy function. - PGH1(Prostaglandin G/H Synthase 1) has possible binding to toxicity targets - SMILES: O1C([C]=[C]C2=[C]C(=C([O])[C]=C12)C1=[C]OC2=[C]C(=[C][C]=C2C1=O)[O])([C])[C] - Compound Name: Puerarone										
4		Low toxicity	Not harmful	Not harmful	Not harmful	Harmful	Not harmful	Harmful	Harmful	Not harmful	Not harmful	Harmful

	 PubChem CID: 14630492	- No possible binding to toxicity targets. - SMILES: O1C([C]=[C]C2=C1[C]=c1oc3c(c1=[C]2)c(=O)oc1c3[C]=[C]C(=[C]1)[O])([C])[C] - Compound Name: Sophoracoumestan A

Parameters included gastrointestinal absorption, blood-brain barrier permeability, and bioavailability score. Compounds with high gastrointestinal absorption, an acceptable bioavailability score, and that follow the Lipinski rule were considered pharmacokinetically favorable compounds. From the top 100 lead compounds, 8 were identified as having favorable pharmacokinetic properties, including high GI absorption, good bioavailability, and drug-likeness filters. 2D interaction analysis shows the types of interactions between the ligand and amino acid residues, which helps identify residues involved in binding and verifies toxicity using the Mcule built-in toxicity checker tool. 4 non- toxic lead compounds to predict the toxicity using ProTox 3.0. Toxicological parameters include oral toxicity, organ toxicity (hepatotoxicity, nephrotoxicity, neurotoxicity, cardiotoxicity, immunotoxicity), carcinogenicity, cytotoxicity, and nutritional toxicity. Based on predictions, lead compounds showing lower oral toxicity and minimal organ level toxicity were identified as promising candidate phytocompounds.

Binding energy (score generated while performing molecular docking) represents a value generated during molecular docking that estimates how strongly NUSAP1 binds with phytochemical compounds. The first complex forms one conventional hydrogen bond with the Glu197 atom and a carbon-hydrogen bond with the Ile199 atom, with a binding energy of -7.084 Kcal/mol. The Corylin complex forms one conventional hydrogen bond with the Arg250 atom, with a binding energy of -6.555 Kcal/mol. The Puerarone complex forms the conventional hydrogen bonds with the Arg250 and Tyr202 atoms, with a binding energy of -6.666 Kcal/mol. The Sophoracoumestan A complex forms no hydrogen bonds, with a binding energy of -6.639 Kcal/mol.

CHAPTER 6: CONCLUSION

Based on literature review studies related to breast cancer, identified commonly reported hub genes, a total of 87 hub genes were identified, 51 of which are known to have therapeutic drugs, and 36 hub genes lack drugs in the Therapeutic Target Database. For 36 hub genes, constructed the PPI networks and identified hub genes using Cytoscape and STRING. By analyzing hub gene expression in breast cancer and pan-cancer using GEPIA2, revealed that the NUSAP1 gene is highly expressed in various types of cancer, and the NUSAP1 gene is selected for further analysis. Using Uniprot variants for NUSAP1 were collected and by using prediction tools for identification of pathogenic variants in Ensemble Variant Effect Predictor, four pathogenic variants were identified: T114N, N386S, K404E, and R434M. All variants (T114N, R434M, K404E, and R434M) were predicted to destabilize the NUSAP1 protein using I-Mutant 2.0 and MUpro; the ΔΔG values are negative, which indicates destabilization. Evaluated the structural impact of nsSNPs on the NUSAP1 variants (T114N, N386S, K404E, and R434M) using the HOPE, which revealed affect the structural stability of the protein. Variants R434 show higher pathogenicity due to alterations in residue charge, hydrophobicity, intermolecular interactions, and MetaRNN score. Wildtype NUSAP1 has a high degree of evolutionary conservation as compared to the R434M variant and the T114N variant. Conservation is reduced particularly in R434M, which suggests it affects protein stability. 3D structural models for wildtype NUSAP1, T114N variant, and R434M variant were generated using AlphaFold, refined with GalaxyRefine, and validated through SAVES. All models have an ERRAT score above 90%, and the Ramachandran favored region occupied above 93%, indicating good stereochemical quality and structural stability. iMODS normal mode analysis confirmed that the wildtype has structure stability and can be used as a receptor for further virtual screening studies. Using PrankWeb identified the binding site for ligand binding. Virtual screening using Mcule identified 4 non- toxic lead compounds for wild-type NUSAP1with good drug-like and low toxicity properties. Again, performed virtual screening with phytochemical compounds, identified 4 non- toxic lead compounds for wild-type NUSAP1with good drug-like and low toxicity properties.

CHAPTER 7: SUGGESTIONS FOR FUTURE WORK

- Molecular dynamics simulations using GROMACS can be performed for wild-type NUSAP1, R434M variant, and T114N variant to extend the current findings, as it gives a realistic view of how the protein moves. Trajectory analysis includes RMSD (Root Mean Square Deviation) to evaluate the overall structural stability of the protein, RMSF (Root Mean Square Fluctuation) to find the flexible regions, and Rg (Radius of Gyration) to evaluate the compactness of the protein.
- Computational models may not fully capture the dynamic complexity of proteins and biological systems; in vivo studies are needed to confirm the results.

CHAPTER 8: REFERENCES

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